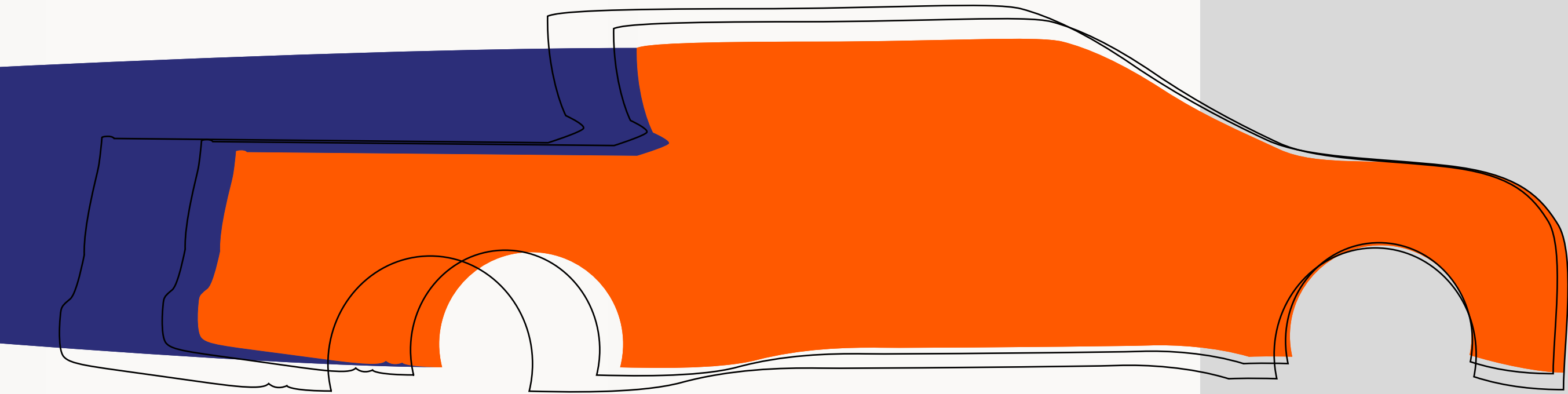


Thermofluids

Group Submission

PAGE 1



Selected Vehicle: **Pickup Truck**

Leila Ali 02584835

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Noah Hett 02558348

Vehicle Summary

DE2 Thermofluids 2 - Part 1

A full sized, crew-cab pickup truck, prioritising long-haul drives and non-utilitarian usage while upholding key capabilities of a pickup truck i.e raised suspension height, full size bed, good approach/departure angles.

Aerodynamic choices made to improve straight line high-speed long journeys.

Design Objectives

C_d ↓
REDUCE DRAG

— C_L
PRODUCE DOWNFORCE

WAKE ↓
MINIMISE WAKE

Key Features

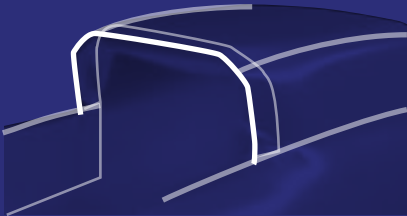
LAYERED PROFILES



The major shape of the truck was derived through optimising the body to fit around each layer while still meeting the required ‘dimensions’ for wheels/arches, bed and cab. From this the connections between the layers was formed.

REAR CAB SPOILER

Rear section of the Cab designed to fill wake left by the relatively flat back of the cab. Air is drawn from sides and top without interference with tailgate or causing turbulence.



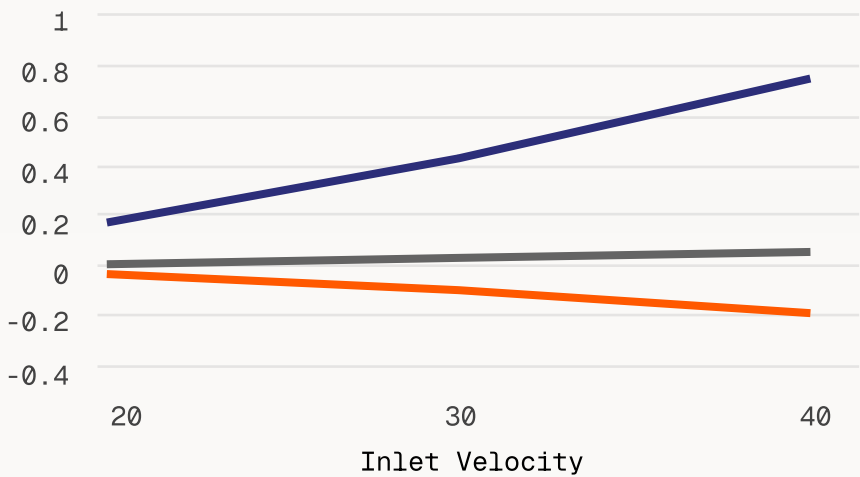
FLOOR PROFILE



Gentle rising floor profile to reduce wake and turbulence at rear of the vehicle.

CFD Findings

Forces and Coefficients



The vehicle was simulated at a range of different wind speeds, re-calculating the forces around the car can indicate different behaviours of major components at these different speeds as well as an ‘optimal’ speed for aerodynamic performance with the current configuration.

C_d 0.44

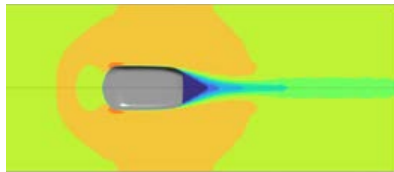
C_{sf} ---

C_L -0.08

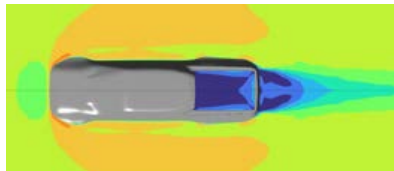
The final coefficient of track of the truck comes to 0.44, stacking up well against competitors such as the Ford 150 - also achieving a 0.44 Cd. However, this design does generate slight downforce, making it more stable at high speed.

Plotting and Visualisation

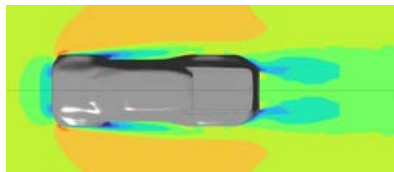
PRESSURE PLOTS



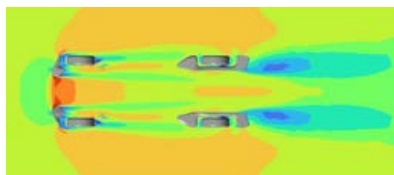
- Clear converging flow from cab spoiler
- No large pressure difference outside of cab rear and ambient



- Converging flow from top of bed
- stable, ambient pressure around wheel arch

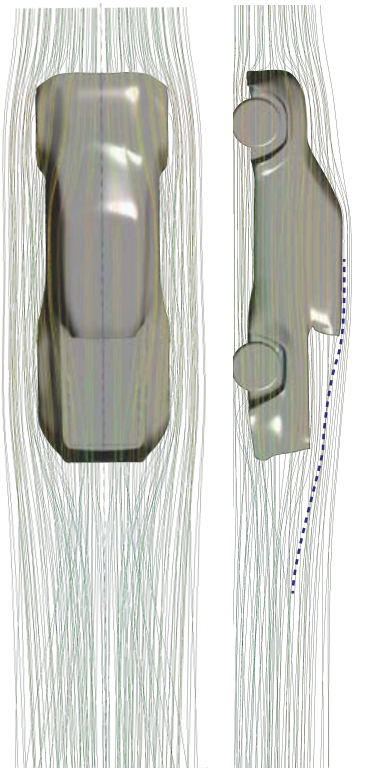


- Converging from lower quarter panel
- Higher pressure from tunnel outlet
- low pressure ‘tips’ at edges of outlet



- Lowest rear pressure at low point behind wheels

STREAMLINE VISUALISATIONS



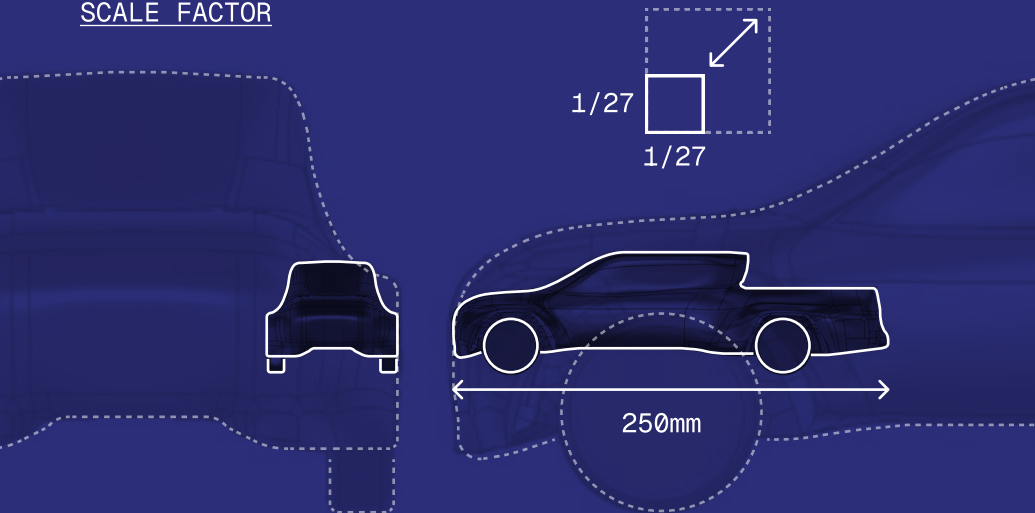
- Stable flow at wheel arches, cab spoiler
- Fewer streamlines at centreline of car
- Potential turbulent points
- Streamlines lowering from top of cab
- General streamline flow from roof to rear
- slight pull-up from tunnel exit for some streamlines
- most streamlines continue straight along length of car
- Streamlines confirmed efficacy of design decisions
- Especially for the flow over the bed which creates a sealing effect.

3D Printing

DE2 Thermofluids - Part 2

CAD Preparation

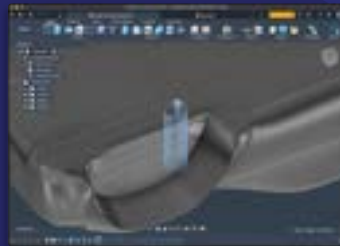
SCALE FACTOR



The model was scaled by 1/27 in lengths compared to original size. This allowed us to maximise size within allowed dimensions, resulting in a length of 250mm.

BOLTS, SMOOTHING AND THREADING

The bolts were added (add dimensions/spacing to match wind tunnel bolt pattern)
the holes were threaded for better compatibility

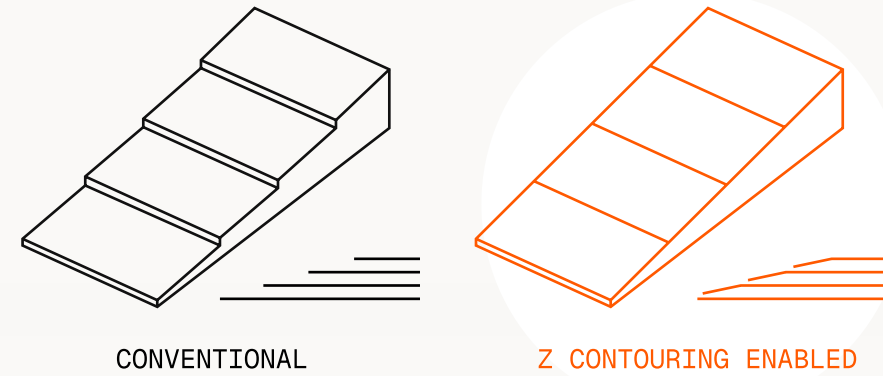


The wheels were Joined to the main body using 'axels', smaller joints in the centre of the wheel, also increasing real-life accuracy.



Printing Methodology

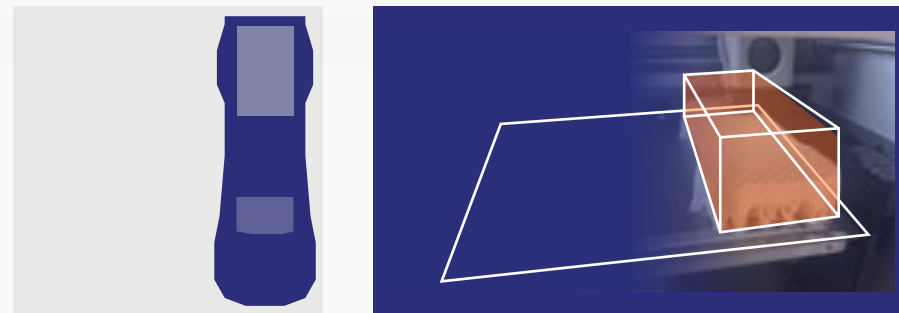
Z ANTI-ALIASING



We used a custom, modified version of Bambu slicer that utilises non-planar top surfaces. This eliminates the stair-stepping effect of print layers. Turning on the 'Z contouring' setting, this has an impact on flow-speed and other slicer settings.

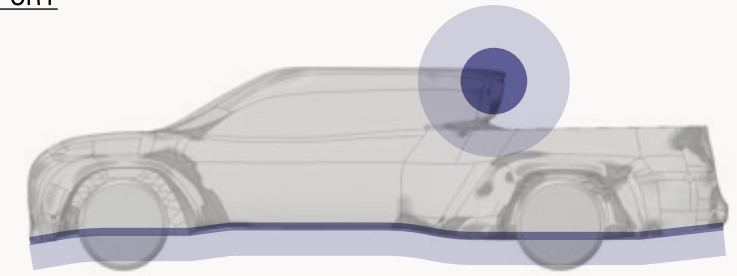
<https://github.com/adob/BambuStudio-ZAA>

ORIENTATION



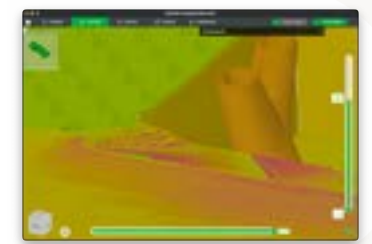
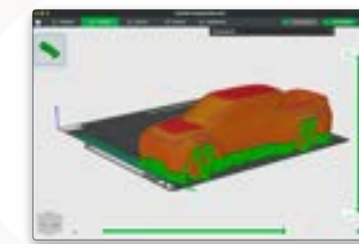
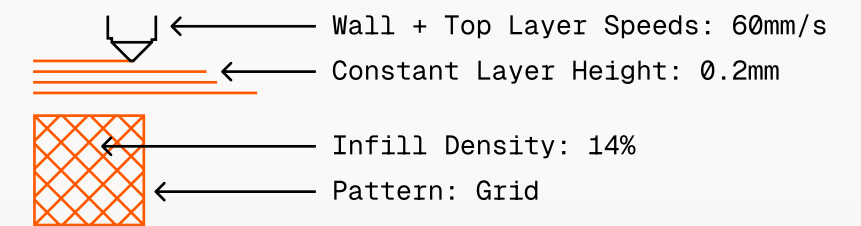
The print was positioned on the left to minimise travel time for print head. Fitting the model front-to-back and vertically un-tilted means we can take full advantage of non-planar top surfaces.

SUPPORT



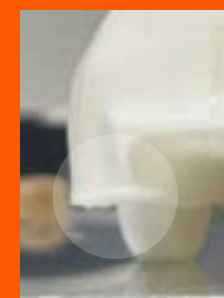
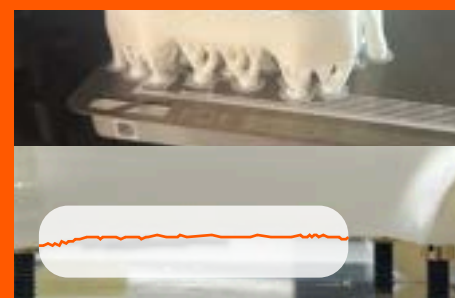
Overhangs, specifically those layered on top of other parts, were difficult to achieve with traditional supports, so tree supports were used, though these created more complexity for the large underside section.

'QUALITY' SETTINGS



We optimised for fast print time while maintaining the outer surface quality, infill is minimal but the slower wall thicknesses and 0.2mm layer maximise performance with ZAA.

Results + Post Processing



Underside surface very rough due to to large area of supports. This is especially bad in areas with higher angle to build-plate or more complex xy geometry



Threaded inserts were then pushed into the Due to us pre-threading the holes, not all inserts seated the same into the model.

Wind Tunnel Setup

DE2 Thermofluids - Part 2

T1 WIND TUNNEL

Subsonic test facility used to experimentally evaluate aerodynamic performance. Provides a consistent airflow over a scaled model, enabling direct comparison with CFD results.

MATCHING CFD CONDITIONS

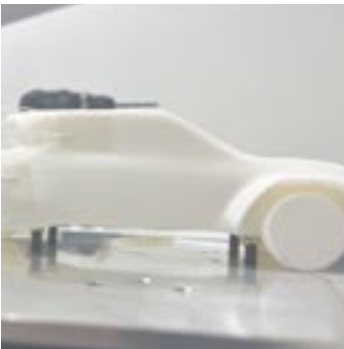
The Reynolds number is expected to be broadly similar due to the small velocity difference (30.5 m/s CFD vs 31.5 m/s wind tunnel), although differing model scales (1/17 CFD vs 1/27 wind tunnel) may introduce some variation. Free-stream velocity is closely matched, giving comparable dynamic pressure and the ground condition is consistent between cases, both using a stationary floor.

Setup Process



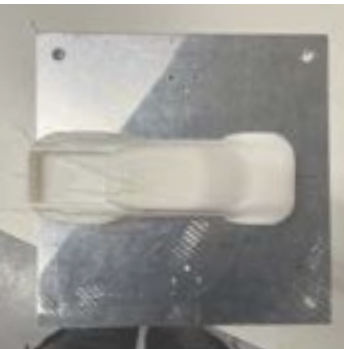
MODEL PREPARATION

Flow visualisation strings are attached to desired spots using tape on the model's surface



ATTACHING BOLTS

Bolts are test-fit into threaded inserts that are fixed to the base of the 3D printed model



FIXING TO PLATE

Model is then bolted to the mounting plate, attached in four points in a square pattern



TUNNEL PLACEMENT

The mounting plate is then transferred into the wind tunnel and bolted to the force detection plate

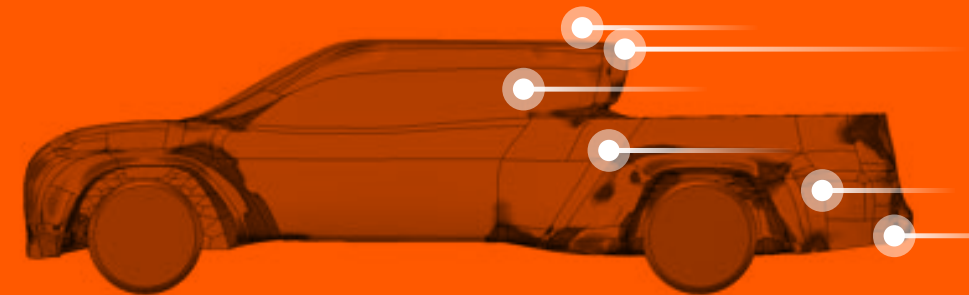


CALIBRATION

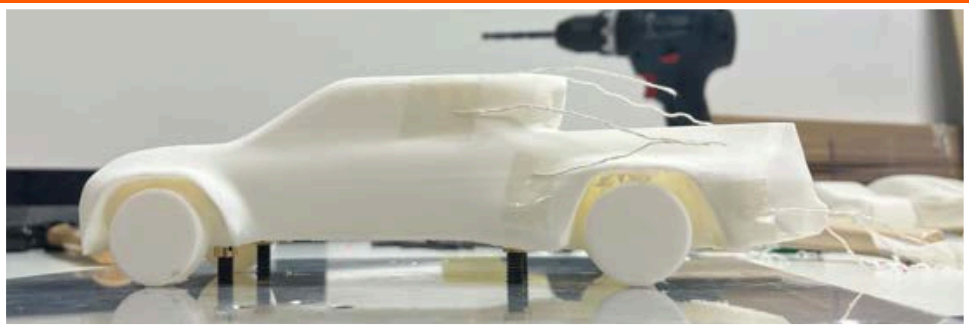
The plate is then calibrated to 0 (tare) before the wind tunnel is activated.

Flow Visualisation

STRING POSITIONING



When positioning our flow visualisation, we focussed on flow at the rear of the vehicle, as this is where the most extreme geometries were located. We used a total of 6 positions, 3 on the side vehicle's side profile and 3 on horizontal surfaces. The roofline has strings of 2 different lengths, one long to measure the 'sealing' effect of the low pressure zone in the bed, and a shorter length to judge flow separation at the cab spoiler



For our real model, we only applied strings along centre line and across one side, this allowed more complete flow visualisation analysis from side-on and top-down angles.

EXPECTED RESULTS



Taking judgement from CFD, we can predict that the side-profile strings will remain laminated to the surface and detach relatively cleanly. For the roofline strings we expect a downwards movement towards the bed, affected by the low pressure zone.

Wind Tunnel Results: Forces

DE2 Thermofluids - Part 2

Data Processing

SELECTION/FILTERING



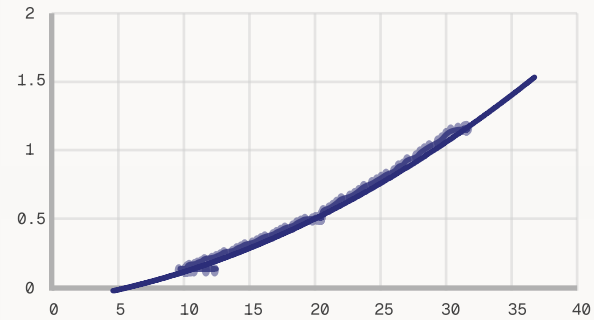
The raw experimental data needed filtering for the more accurate calculations.
Relative Velocity Check: comparing air speed value vs its previous to ensure stability and consistency.
Speed Binning: grouping values into 0.5m/s ranges to separate out 10.5, 20.5 and 30.5 for calculation.

CALCULATION

coeff. calculations only use values in correct 'bins' + with consistent velocities. Despite 'binning' more exact air-speed values were also calculated e.g 10.32 for 10.5 bin.

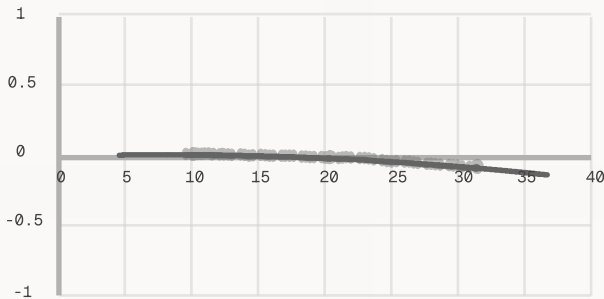
Forces and Coefficients

DRAG FORCE



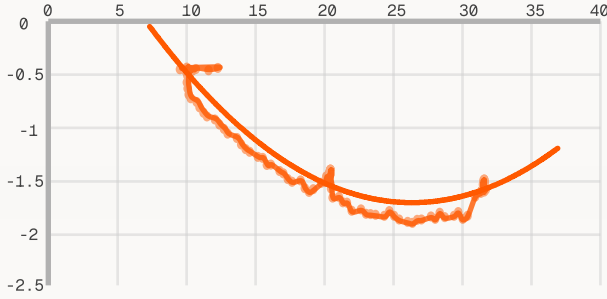
All reported side forces have low magnitudes, never reaching above 0.1.
Despite this small value, we do see side forces decreasing as air speed increases.

SIDE FORCES



All reported side forces have low magnitudes, never reaching above 0.1.
Despite this small value, we do see side forces decreasing as air speed increases.

LIFT FORCE



Recorded lift force sees extreme deviation and non-linearity. it starts near 0, with a large decrease before rising again. The measurement also 'jumps' at 10.5, 20.5 and 31.5 due to flow-settling.

	10.5	20.5	31.5	CFD(30.5)
Cd	0.519	0.506	0.482	0.44

Drag Coefficient remains both consistent and close to our predicted values.

It sees a slight decrease in value at speed increases, dropping 0.037 from 10.5 to 31.5 m/s air velocity.

	10.5	20.5	31.5	CFD(30.5)
Csf	0.038	-0.011	-0.032	---

The increase in side forces with air speed is reflected in our Csf
Due to small magnitude of the measured side forces, we see a large percentage change in Csf (60-120%).
However, this is not a major problem as the numeric changes in coefficient never top 0.07.

	10.5	20.5	31.5	CFD(30.5)
Cl	-1.762	-1.406	-0.641	-0.08

The lift coefficient we achieved sees very extreme variation. Even at it's lowest magnitude, the value is unrealistic and completely incomparable to the CFD simulated result.

*Unrealistic Result, potentially due to sensor read error.

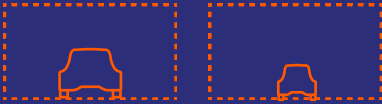
CFD Disparities

DRAG

The disparity between the wind tunnel experiment and the CFD predictions is 0.042 [at an air speed of 30.5-31.5] this is a fairly significant difference and has many potential causes.

BLOCKAGE RATIO

CFD: Wind Tunnel:
increased ratio will increase drag



SKIN FRICTION

Non-uniform surfaces of 3D printed model cause greater surface friction, increasing the skin-friction element of drag.
This also helps to explain the decrease in Cd with speed: as reynolds number increases, the boundary layer is more turbulent. The air 'sticks' to the car for longer, meaning that the wake is smaller, reducing drag

OBSTRUCTIONS

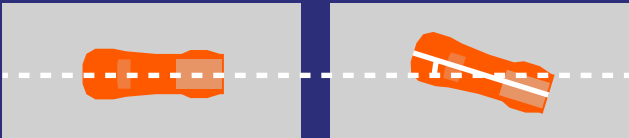
Compared to the 'perfect' CFD in CAD, the wind tunnel has extruding bolts, increasing frontal area and disrupting airflow on the vehicle's underside.

SIDE FORCES

The main difference between side forces in CFD and practical implementations was that the magnitudes of these forces were increasing as air speed increased.

MODEL TILT

The primary explanation for this is that the wind tunnel model was tilted, causing air to hit the side of the vehicle, therefore introducing side forces.



LIFT

Assuming that this is not due to a sensor reading error then it would be something to do with changing flow paths with speed. At different air-speeds, detachment points will be varied, also having an effect on the wake behind the truck.

MEASUREMENT SENSITIVITY

Lift force measurement is the most sensitive, it can be seriously effected by underbody flow paths and pitch angles.

AIRFLOW SETTling

Due to the extremely short testing period, the time given for airflow to settle was only a few seconds per speed. This may have impacts on the flow, particularly under the car.

MOUNTING ISSUES

Due to the extremely short testing period, the time given for airflow to settle was only a few seconds per speed. This may have impacts on the flow, particularly under the car.

Wind Tunnel Results: Flow

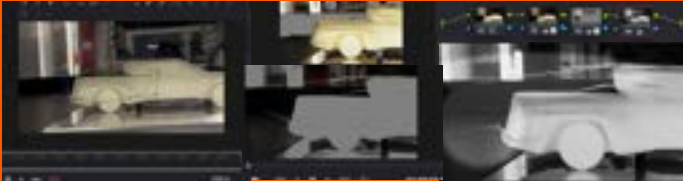
DE2 Thermofluids - Part 2

Processing

we wanted to analyse the turbulence of the string

- could not use fourier on the freq as too low fps
- using edge detection algorithm
- model string as a straight line to track angle/deviation
- problems with initial method due to white-on-white tracking

VIDEO PRE-PROCESSING



1. Stabilised, isolating any movement or shift in perspective from the video
2. Increased global contrast for midtones
3. Darkened + blurred background, using selections around the vehicle.
4. Keyed - for brightness, changing thresholds to just catch the tufts
5. Contrast and Midtone Detail increased

SCRIPTING

Setup

Tracker is setup for each string using a clear frame of video. Each tuft is defined with anchor + end point and a ROI, (region of interest).

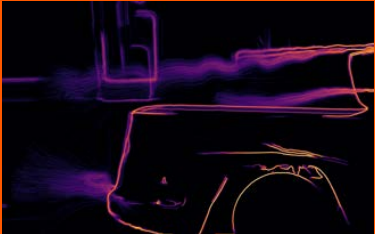
Tracking

Each frame is computed within the ROI. Local contrast is boosted then edge-detection is run. The dominant axis of edges is found using a 'PCA-like operation'

Quality Control

Only around 40% of frames were usable due to blur. To choose, the linearity of the point-cloud of edges is checked.

RESULTS



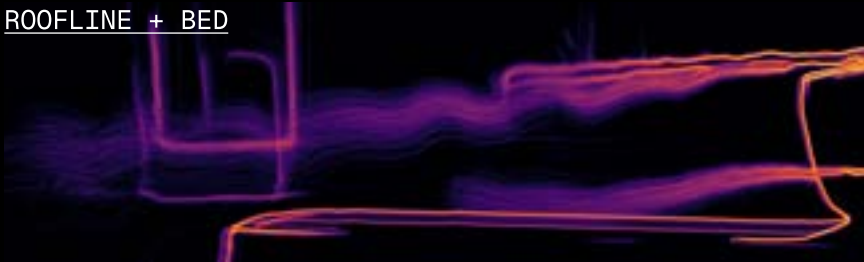
Results

Line approximations are accumulated, then normalised and coloured as a heat-map. We see clear, lines proving effective tracking.

Analysis

We focussed on flow at the rear of the vehicle, as this is where the most extreme geometries and therefore differences to CFD would occur. This analysis was split this up into 2 physical areas, the wake from the bumper/fender section and the effect of air moving over the bed.

ROOFLINE + BED



SIDE-CAB shows good flow separation, it gets sucked behind the back of the cab. This is visibly consistent, shown by the defined shape in the visualiser.



SHORT remained super stable and pointing slightly down, exactly what was intended and predicted. LONG was not as effective but did prove that there is an overshoot of the bed. It stayed vertically clear, even with the effects of mass.



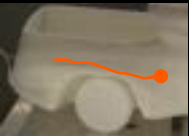
FENDER + BUMPER



Bottom-attached string had significant turbulence, and even reversed flow. Separation in this area is poor. This is a contrast to the attached to the rear side-bumper, despite being on the same 'level' vertically it remained still, with a slight uplift.



Wheel-arch string was not able to be tracked, though on visual inspeciton, it did not move at all, staying stable and laminated to the vehicle's surface.



CFD Disparities

ROOFLINE + BED

Turbulence from behind the bed is not found in our wind-tunnel case. This could be due to a lack of strings in the upper part of this region. Primarily the LONG string being too high, overshooting the bed and settling above the predicted turbulent area.



FENDER + BUMPER

The string attached to the bottom of the diffuser showed a much greater turbulent flow separation. This caused disruptions to the wake that were not found in CFD analysis. A likely contributor to this is the mounting bolts, though some turbulence could be mitigated with improved diffuser geometry.

Design Optimisation

DE2 Thermofluids - Part 2

Results Reflection

FORCES + COEFFICIENTS

The most clear and measurable distinction found in our wind tunnel analysis is that drag increased compared to CFD predictions by 0.037. This raises it to 0.482 @ an air velocity of 31.5m/s, now underperforming against competitors (Ford F150 - 0.44). Side-force disparities were considerably smaller, and lift was incomparable, so our main focus will be drag reduction.

FLOW

Analysis of flow identified the main points of increased drag at the rear of the vehicle:
The underside - where we see the greatest site of turbulence and poor flow separation.
The 'overshoot' of flow over the bed - where it remains a considerable distance above the top of the tailgate.

Problem Areas

DIFFUSER

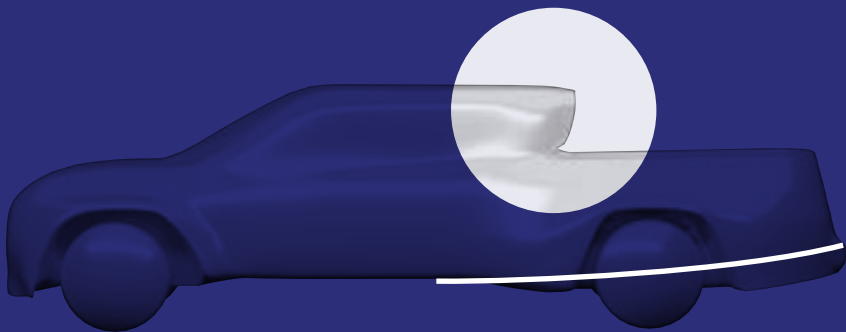
the turbulence is happening low, interfering with the flow of the diffuser, compounding with the turbulent wake of the vehicle.
Increasing the diffuser angle allows greater expansion of air, especially in the upwards direction. This aims to reduce the amount of rear wake area and encourages a more upwards flow path. However, its important that the rear fender geometry is not changed as wind tunnel testing validated the effective attached flow + minimal losses in this region.



ROOFLINE

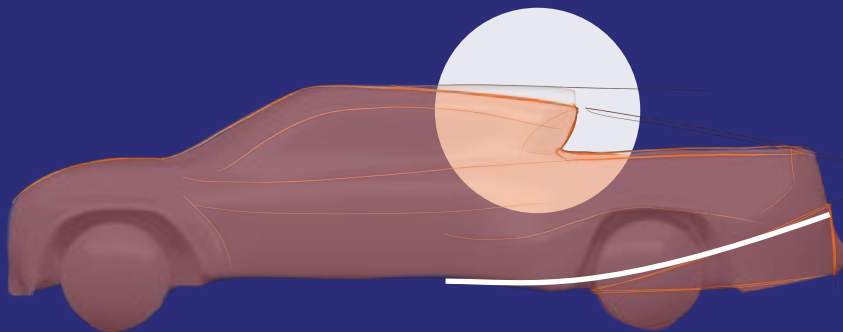
The overshoot increases the size of the wake above the bed. To combat this issue we can increase the slope of the cab, streamlining or directing the flow further downwards over the bed. This is a more effective solution than decreasing cabin height, as interior space and visibility will be better preserved.
It is important that airflow is not directed into the bed, causing a turbulent low velocity system in that area.

Geometry Changes



STANDARD GEOMETRY

The original model includes a slight curve at the end of a flat roofline as well as a small diffuser with shallow angle and depth.

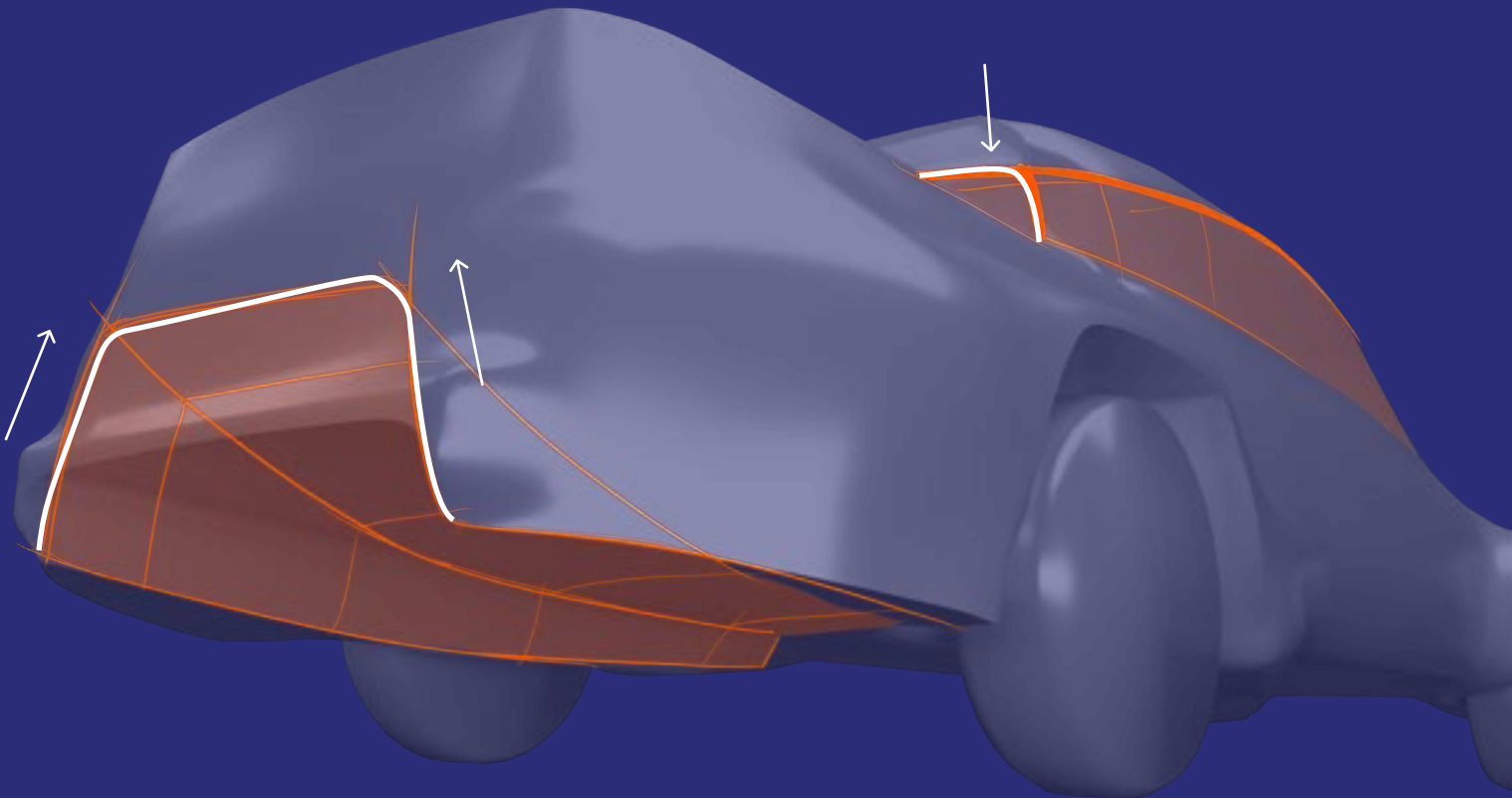


OPTIMISATION RECOMMENDATIONS

The optimised geometry features a lower curving roofline and a more aggressive diffuser configuration.

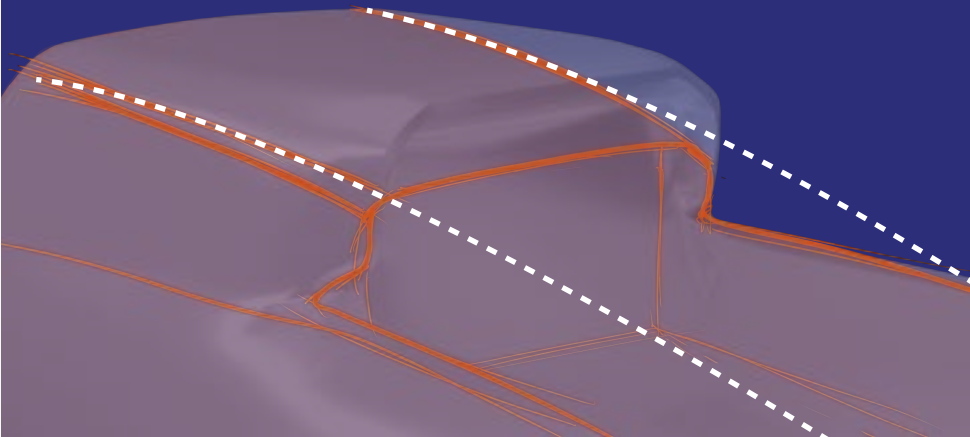
DIFFUSER

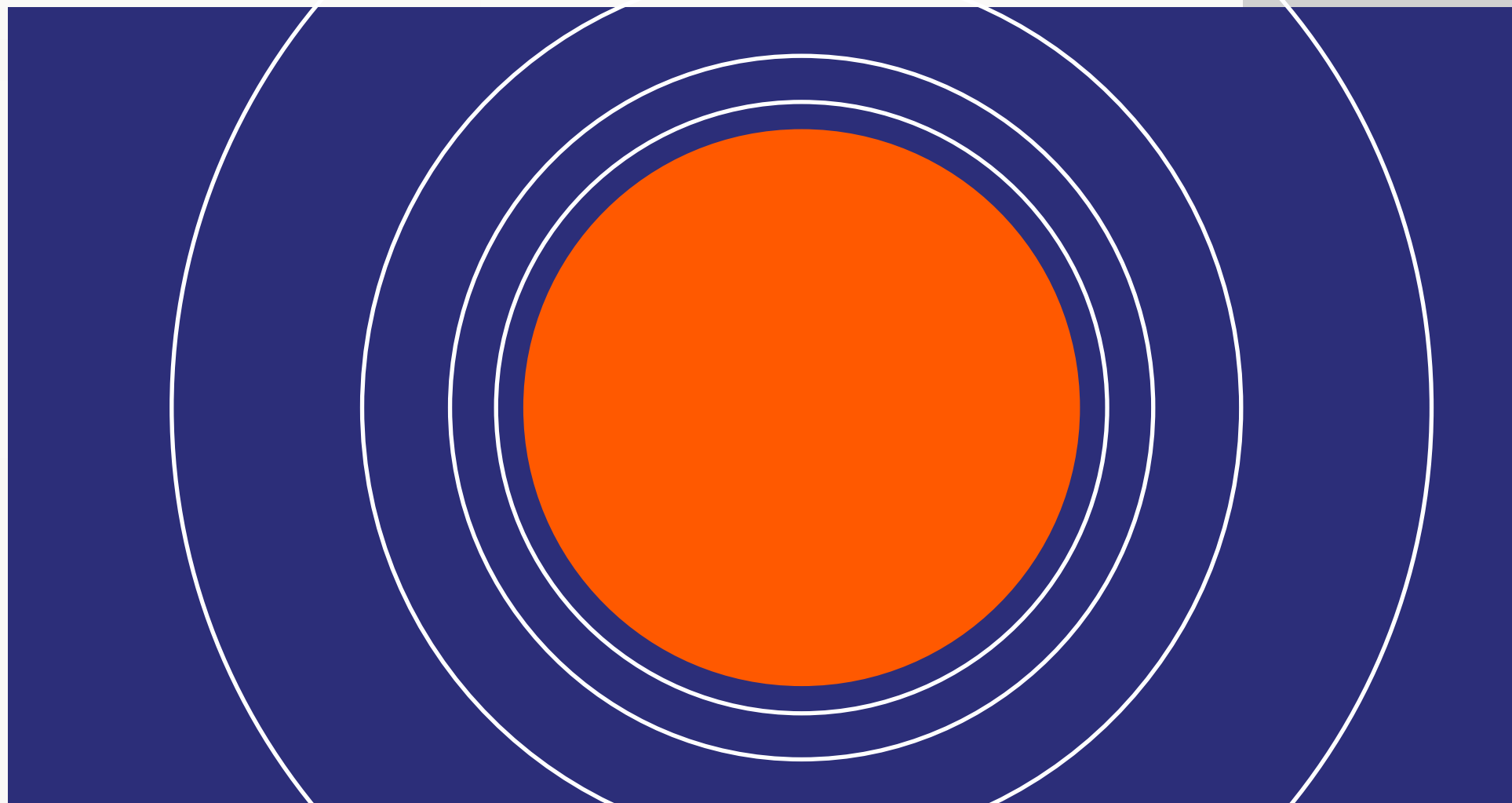
The changes to the diffuser geometry are increased angle and vertical height. Importantly, the sideways expansion remains unchanged, keeping intact the sharp separation surfaces which prevent 'wrap around' effect of attached flow at the sides of the vehicle.



ROOFLINE

The new roofline aligns directly from the top of the roof to the end of the bed. Keeping airflow high enough to guarantee the seal (air 'curtain') over the bed, while minimising the effects of wake vertically above the tailgate.
The side elements and overhang keep their original geometry, allowing air to in-wash over the bed, adding to the sealing effect.

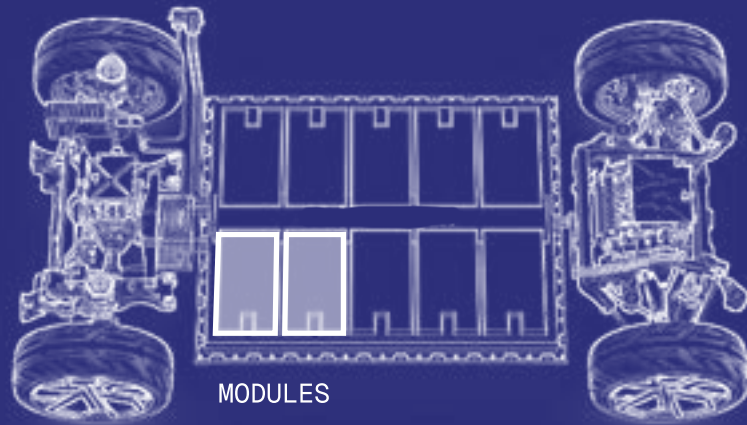




Pipe Fluidics and **Cell Cooling**

Design Summary

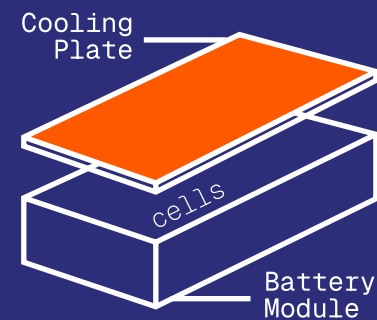
DE2 Thermofluids - Part 2



[1]

Construction + Layout

Battery Electric vehicles are most commonly constructed on a single battery pack. The layout is often a 'skateboard' platform like VW's MEB (above) which underpins 19 vehicles. These large battery packs are segmented into 'modules' of cells.



Design Objectives

PRIMARY OBJECTIVE



Reduce differences in temperature, improving uniformity across the module's surface.

SECONDARY OBJECTIVE



Increase heat transfer, increasing the rate that energy can be transferred from the battery's surface.

PRACTICAL CONSIDERATIONS

From the Module:

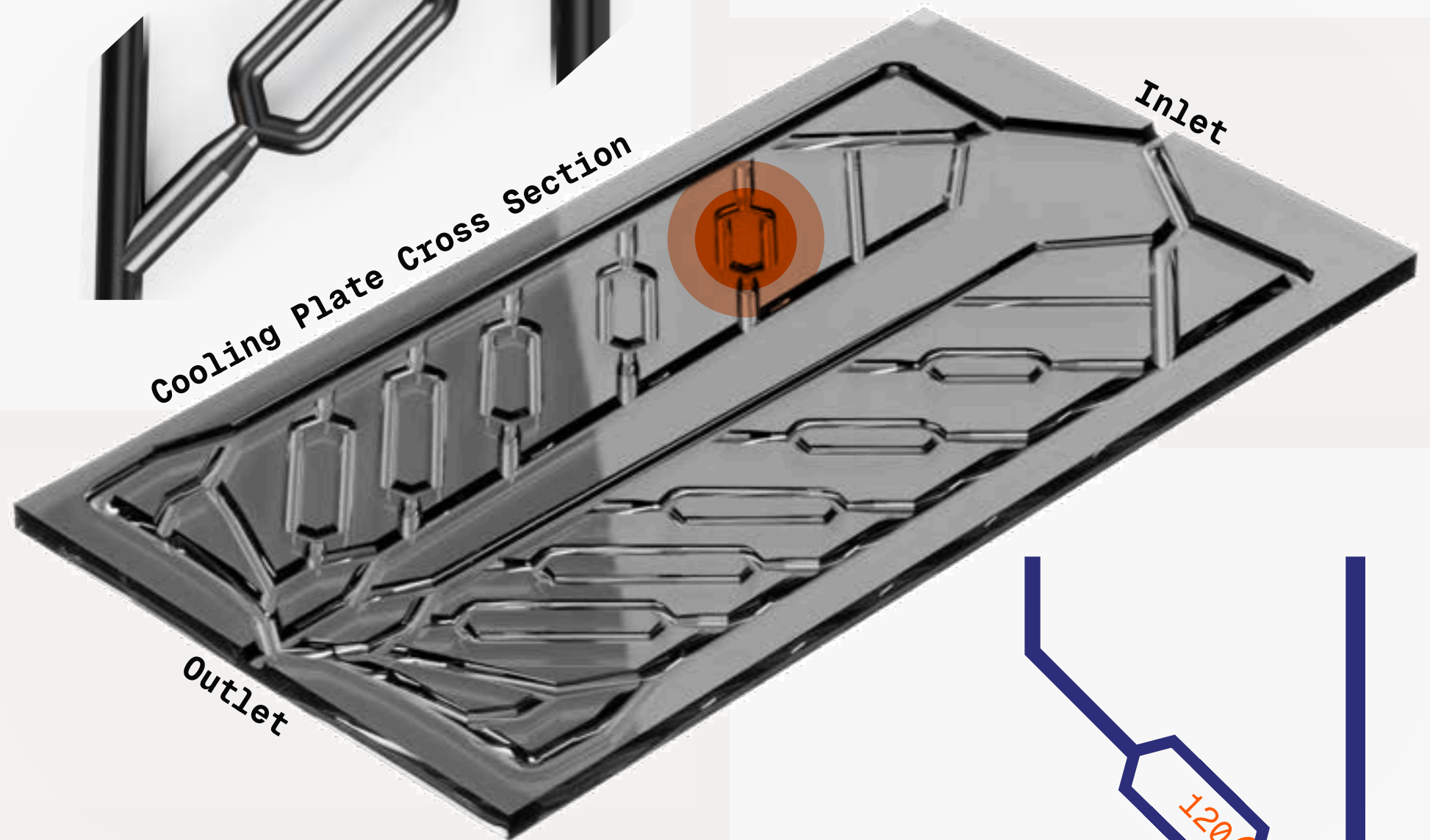
- Single inlet + outlet system
- 440*230*20 max cooling plate size

Other

- Simple cross sectional geometry for manufacturing
- Single-surface cooling plate design

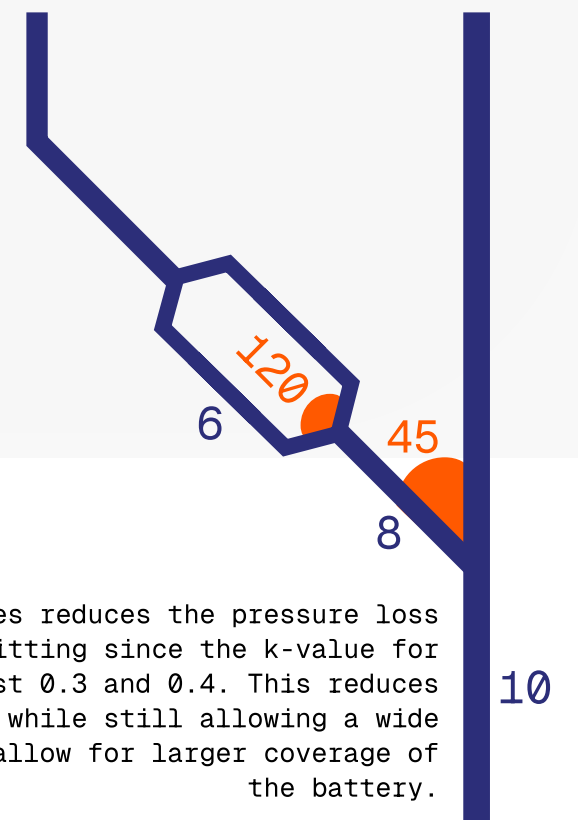
Multiple nested parallel sections allow for a wider coverage of the battery and better heat transfer. This incurs a slight cost to the pressure but since splits are a minor loss it is not a significant issue.

Battery designed inspired by animal wing vein structures which act similarly in how they must distribute blood throughout the wing without excessive increases in pressure. Biomimicry allows the centuries of evolution to be used to guide more modern and mechanical structures. This comes at a cost to manufacturability which may be difficult for this design however just 3 diameters used to ease this problem.



Increasing the concentration of pipe towards the outlet accounts for the fluid heating up as it moves down the battery. Increasing the concentration at the end maintains a more equal heat transfer throughout the plate. The paths then converge onto the single outlet with minimal tight bends to preserve the pressure.

Prioritises efficient and effective heat transfer over minimising pressure losses with many changes in diameter and joining pipes.

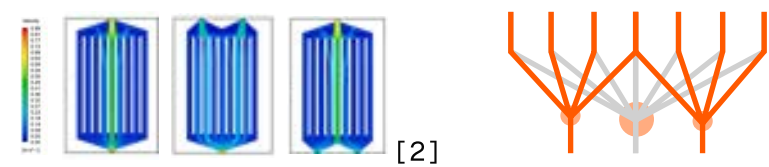


Using these angles reduces the pressure loss when turning or splitting since the k-value for these splits are just 0.3 and 0.4. This reduces pressure losses while still allowing a wide enough angle to allow for larger coverage of the battery.

Benchmarking + Ideation

DE2 Thermofluids - Part 2

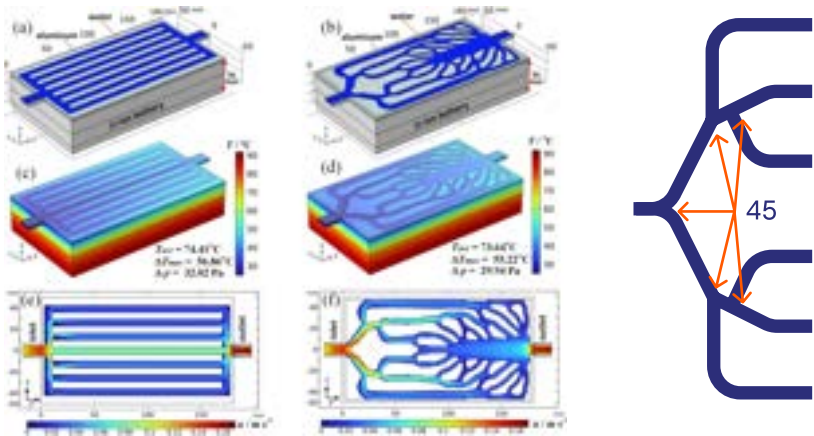
MULTI-INLET/OUTLET



DOUBLE OUTLET

The main benefit of the double outlet configuration is reducing the stagnation and turbulence that combining many flows into 1 causes. By separating this out into 2 outlets, a more constant velocity is achieved, though this will increase pressure and both manufacturing + flow path complexity.

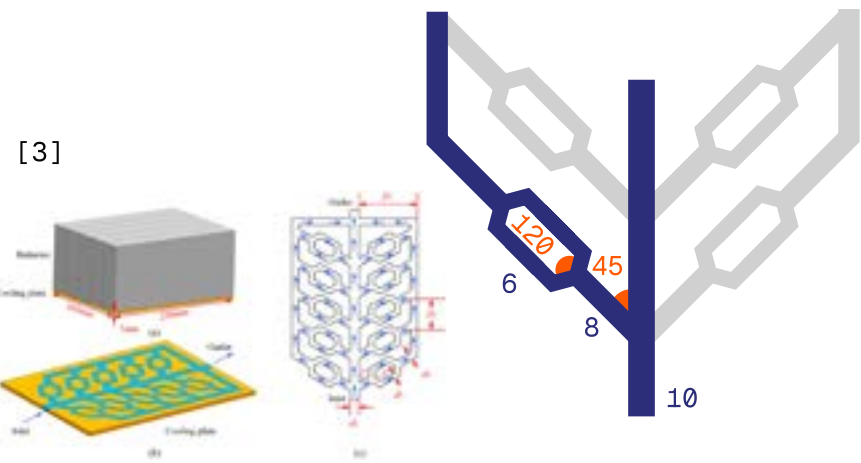
TOPOLOGY OPTIMISATION



INITIAL SPLITTING

This pipe layout was generated based on optimisations for temperature uniformity and pipe pressure dynamics. It successfully reduced maximum temps by over 1C while reducing pressure differences by over 2Pa. The most interesting part of the design is the initial splitting which only uses 45deg angle splits.

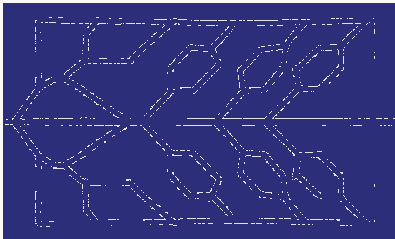
BIO-INSPIRED WING-VEIN



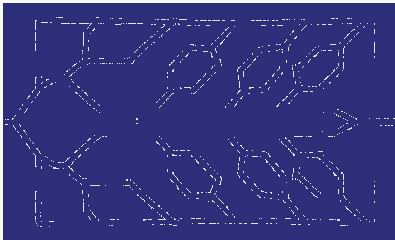
'POLYGON' SIDE CHANNELS

Inspired by vein structures in animal wings, these branches in side pipes aim to better distribute coolant over the surface without a large increase in pipe pressure. From this research it was found that the optimal angles were 45deg for the main branches, with 120deg separating the polygon sections.

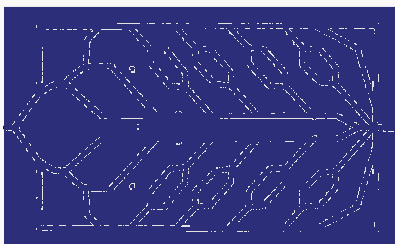
1 initial layout



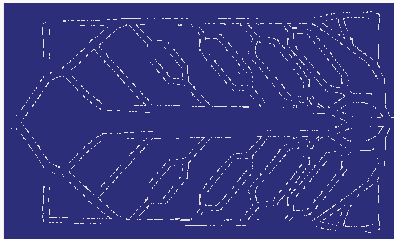
2 Removed Direct Flow



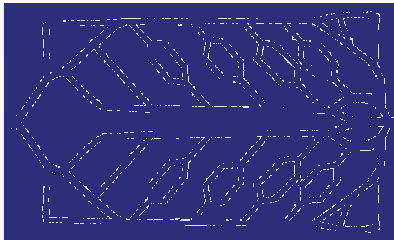
3 New Vein Direction



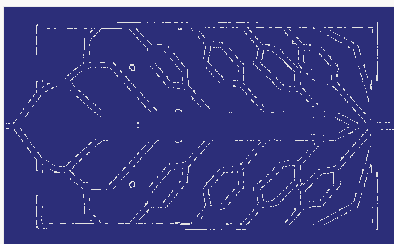
6 Scaling Vein Length



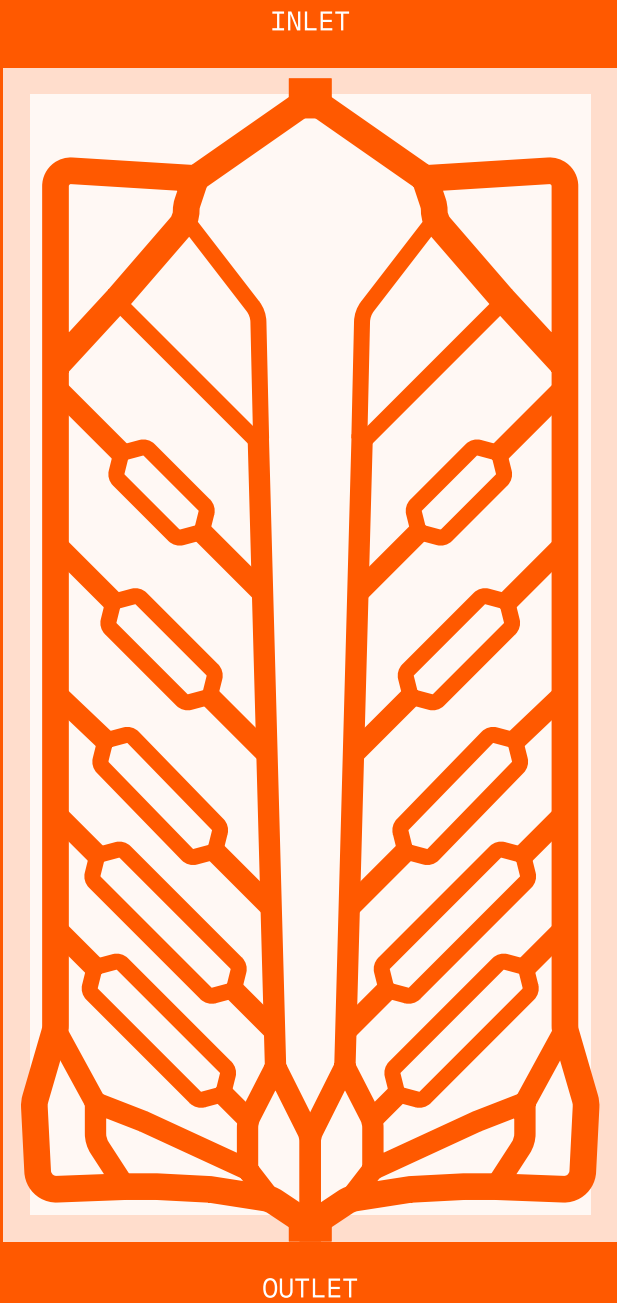
5 Split Outlet Join



4 Better Vein Distr.



FINAL PIPE LAYOUT



- Pipe size limited to 3 specific diameters (10mm, 8mm, 6mm) and corner radii (3mm, 10mm) for manufacturing simplicity.
- Only 1 point of convergence with more than 2 pipes, to reduce complexity and flow stagnation.
- Little to no 'direct flow' or bypass flow-paths, increasing cooling coverage.

Parametrised Process

DE2 Thermofluids - Part 2

Parameterisation Approach

Three internal pipe diameters defined as driving variables across both CAD and hand calculations, allowing geometry and analysis to update in sync between initial and optimised CFD iterations. Using the pipe tool and parameter function on Fusion 360 allows the pipes to be set to variable diameters.

User Parameters			
D1	mm	6 mm	6.00
D2	mm	8 mm	8.00
D3	mm	10 mm	10.00
A0	mm	D2	8.00
A1	mm	D3	10.00
A2	mm	D3	10.00
A3	mm	D2	8.00
A4	mm	D1	6.00
A5	mm	D2	8.00
B1	mm	D1	6.00
B2	mm	D2	8.00
B3	mm	D2	8.00
B4	mm	D3	10.00
C1	mm	D1	6.00
C2	mm	D2	8.00
C3	mm	D2	8.00
C4	mm	D3	10.00
C5	mm	D2	8.00

3 Set Diameters

Reduce manufacturing costs with just three diameters which can be changed to edit all pipes simultaneously.

Pipe Lengths

Assigning different pipe areas different variables which are then set to the diameters allows significantly more control than using Ansys Discovery for parameterisation which selects areas based on diameter, limiting the parts which can be changed.



Hand Calculations

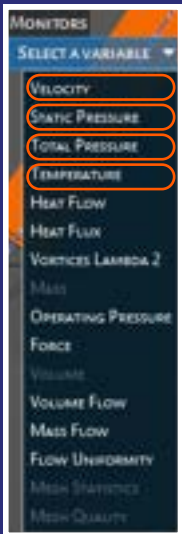
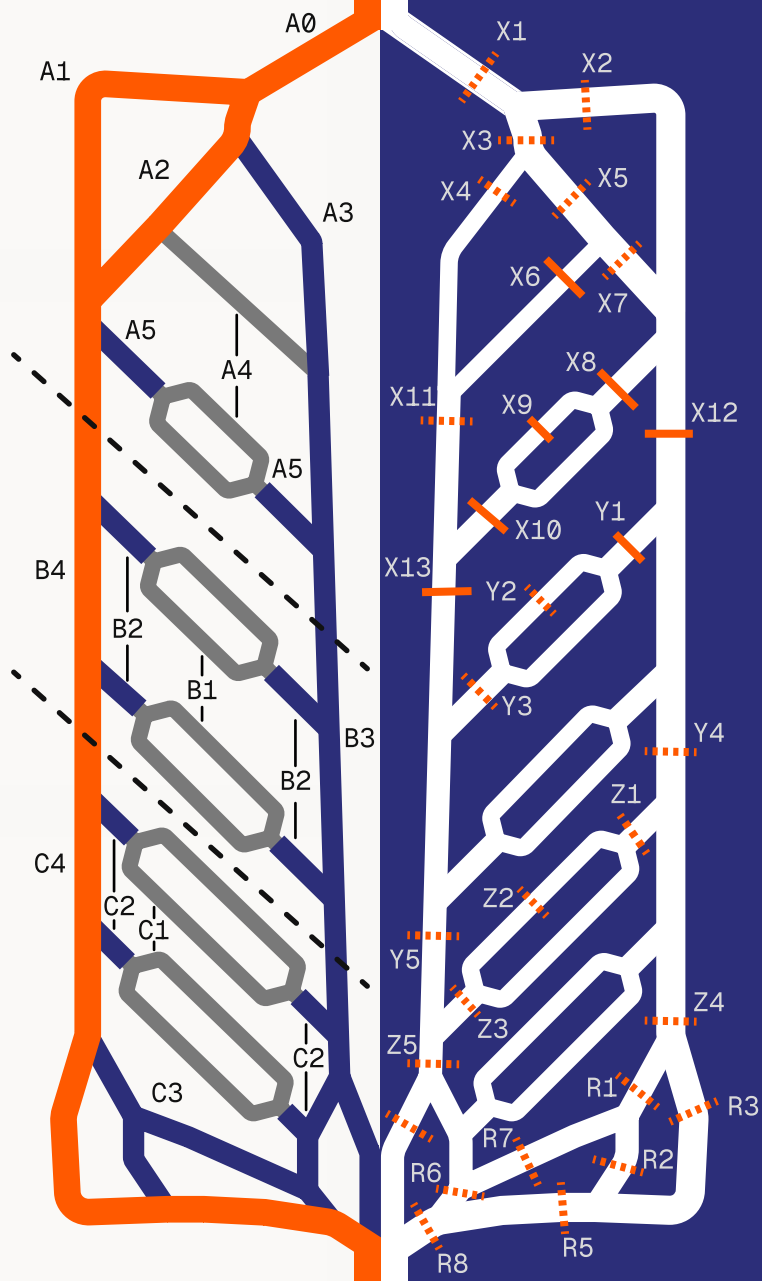
Mirroring how the CAD is parametrised, the same three diameter variables that drive the Fusion sweeps are used within a python script as the inputs for the hand calculation methods.

A main file is used to input pipe flow path and run associated methods, following path structure, varied diameters and junctions. This main file calls a function file to execute each calculation sequentially.

All thermofluids methods (Reynolds number, friction factor, Nusselt number, pressure drop, heat removed) are written as functions that accept diameter, length, and mass flow rate as direct inputs. A single change to a diameter propagates instantly through every dependent calculation without rewriting any logic

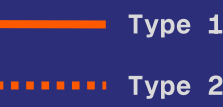
Monitoring

Figure 3. illustrates the intended locations for monitoring points to be defined in ANSYS. Located at key positions along the flow path to track pressure losses and extract local heat transfer results during the CFD run. This allows direct comparison against the hand calculation outputs at equivalent path positions



Two Monitoring Point Types

1. Standard monitoring array captures velocity, total pressure, and temperature to define the flow's energy profile.
2. Points add static pressure measurements, specifically at a single vein split and contraction. To be used in later K-value investigations.



CFD Setup

DE2 Thermofluids - Part 2

System assumptions:

- Battery modelled as a single homogeneous solid
- Contact thermal resistance between battery and plate have been neglected to focus calculations on channel design and heat removal.

1. Creating the Geometry

Geometry of battery cooling plate, battery, and channel were created in Fusion 360 using pipe tool and the combine tool to cut the channel from the plate. It was then imported into ANSYS Discovery.



2. Enclosed Volume

Fluid volume is extracted from the cooling channels inside of the plate to define the region where coolant exists.

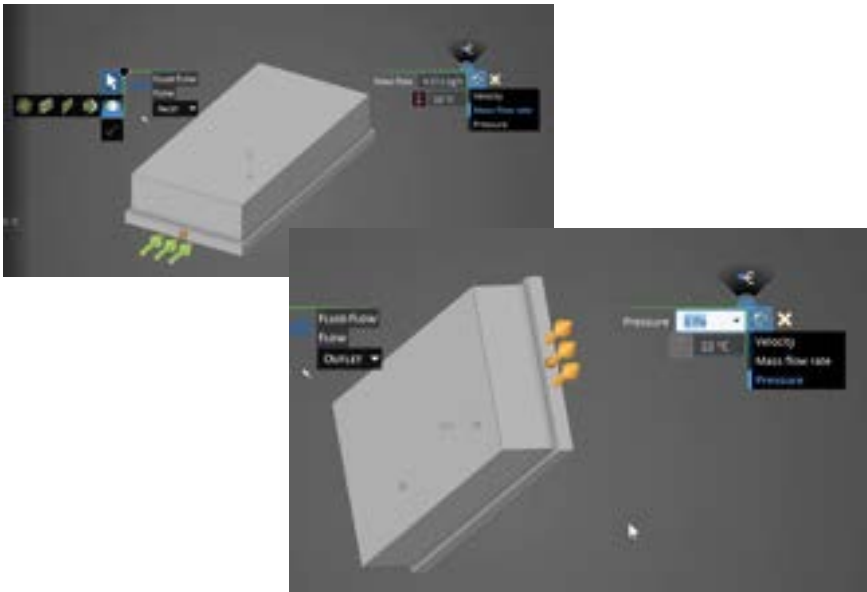
This creates a separate domain for the fluid allowing the software to simulate fluid flow and heat transfer interactions.



3. Inlet and Outlet Flow

A mass inlet flow is applied (10g/s at 20°C) to control how much coolant enters the system. Our numerous junctions and 120 degree bends will lead to more significant minor pressure losses at higher velocities/turbulence. By using a relatively low inlet velocity, we effectively reduce the risk of flow maldistribution. Setting the temperature to lower than the battery temperature enables cooling,

Pressure outlet is set to 0Pa.



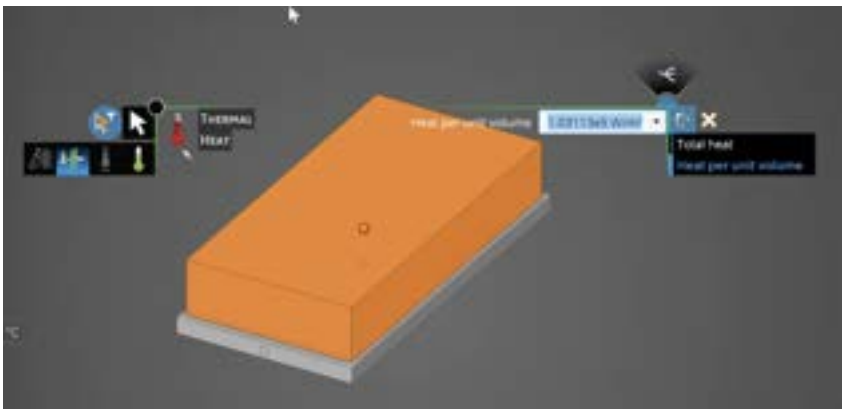
4. Solid Thermal Heat of Battery

Battery modelled as a heat-generating solid

Constant heat generation rate applied to simulate discharge conditions

Heat transfers from the battery to the cooling plate and fluid.

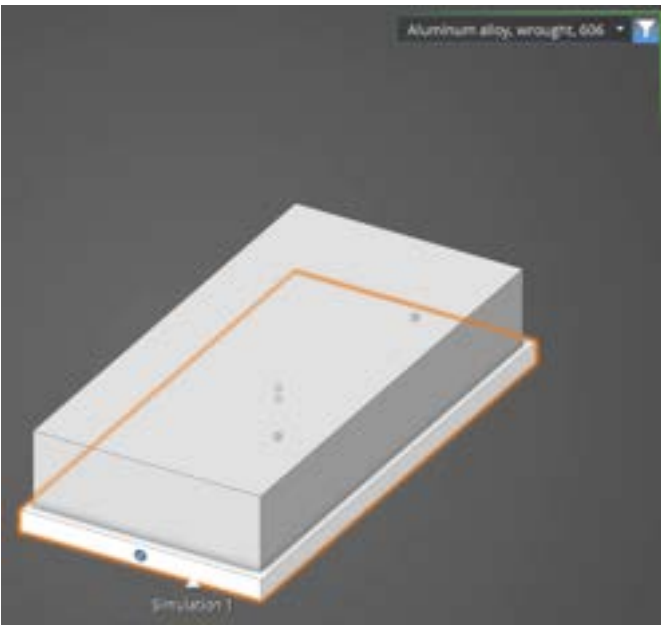
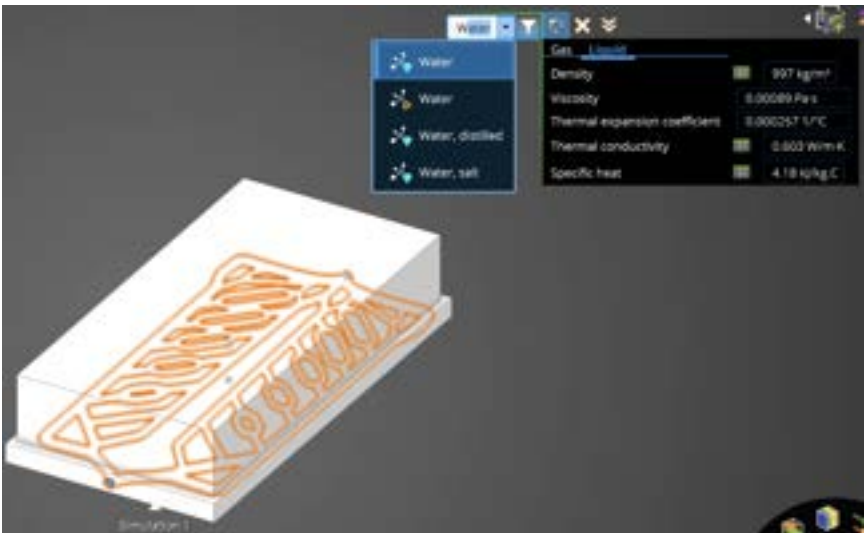
Battery module assigned heat generation rate of 1.03113e5 W/m³



5. Setting Material Properties

- **Coolant:** Water
- **Battery:** 2530 kg/m³, 30 W/m·K 1000J/kg·K
- **Cooling plate:** Aluminum alloy, wrought, 6061, T4.

These properties conduct how the heat is conducted and removed. Assigning aluminium to the cooling plate improves its properties due to the high conductivity facilitating heat transfer. Using water for the coolant is ideal since it is accessible and has a high heat capacity.

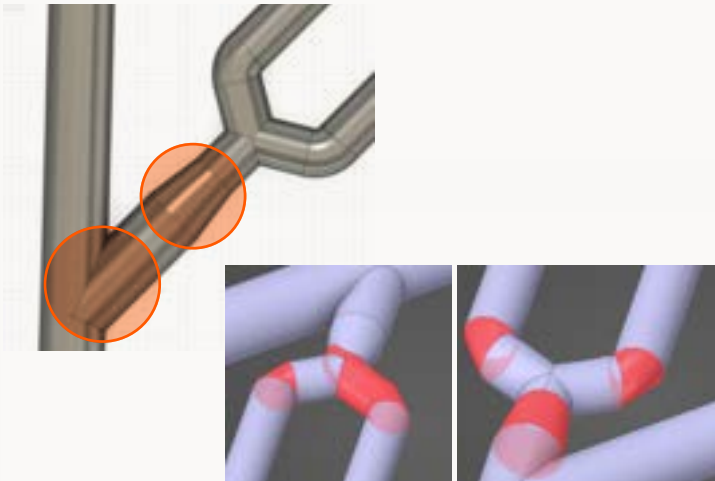


Meshing

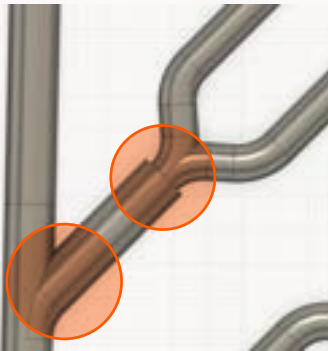
DE2 Thermofluids - Part 2

Changing Geometry

Before



After



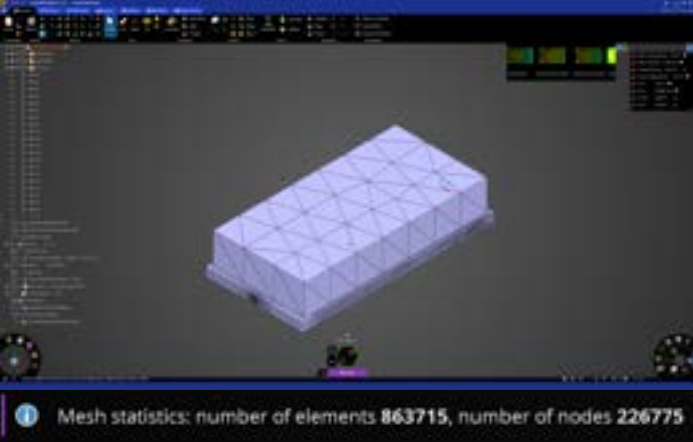
These were amended in the next iteration with lines smoothly parting and coming together. The change away from the chamfered edge as the diameter decreases was not the design however since that is how the hand calculations are calculated, it means the results are more comparable than with the old geometry.

First iterations of CAD couldn't mesh due to sharp corners in the geometry and smoothly changing diameters causing issues.

Meshing divides the entire geometry (solid and fluid regions) into many small discrete elements (cells).

The solver performs calculations on each cell to determine temperature, velocity, and pressure distributions.

This is necessary because the governing equations (fluid flow and heat transfer) cannot be solved analytically for complex geometries, so they are approximated numerically.

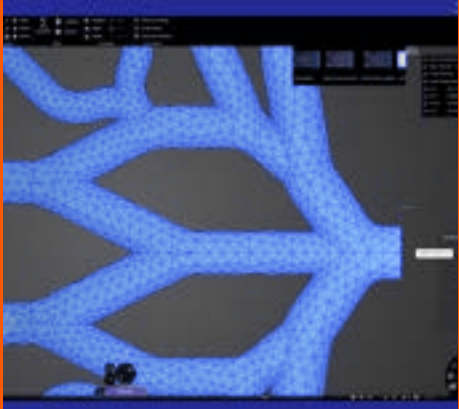


Local mesh sizing allows specific regions to have a finer mesh than others.

- This is especially important in areas with:
- Cooling channels (fluid flow regions)
 - Solid-fluid interfaces
 - High temperature gradients near the battery

By refining the mesh locally, the simulation can more accurately capture heat transfer and flow behaviour where it matters most.

This improves accuracy without significantly increasing overall computation time.



Set up & naming conventions

Each branch/split was assigned a monitor that would measure; pressure, temperature and velocity. A naming convention was used to separate the battery into different sectors responding to the CAD model parameter setup.

Each monitoring point has been selected and brought down to the pipes midway point. Keeping them all level and avoiding boundary layer interference.

By using fixed coordinates, we can vary pipe diameters during our development, while monitors stay in the same spot, making the a fair comparison, provided the underlying geometry remains unchanged.

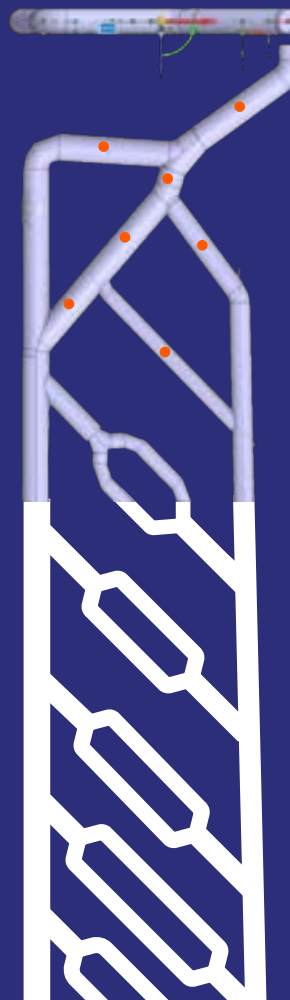
Results Synthesis

Point	Pressure	Temperature	Velocity	Static Pressure
R1	18.73993	22.04128	0.050388	
R10	17.26018	21.87349	0.079625	
R12	17.26279	22.23159	0.17503	
R2	18.09259	21.56204	0.044294	
R3	18.60809	22.31437	0.0111	
R4	17.58578	22.21335	0.023946	
R5	17.78597	21.72408	0.049649	
R6	17.33489	22.0429	0.052532	
R7	17.50595	21.58422	0.014309	
R8	17.5835	22.49375	0.015461	
R9	17.69815	22.5845	0.072929	
R1	27.54507	20.5668	0.088471	27.545067
R10	20.99563	22.87307	0.04526	20.986637
R11	22.01312	23.74557	0.036324	21.33638
R12	23.67467	21.77572	0.077887	20.63374
R13	21.04262	22.75884	0.040778	
R2	23.77835	22.52432	0.023041	23.47648
R3	26.39694	21.4248	0.086933	22.406139
R4	22.69652	22.98034	0.021713	
R5	26.58571	0.056631	20.92953	
R6	22.94818	23.77459	0.027626	
R7	25.50695	21.21047	25.50695	
R8	21.12052	24.44592	0.005963	21.094894
R9	21.05295	23.57427	0.003851	21.045349
R1	20.26474	23.7369	0.007955	20.169729
R2	20.5138	23.41146	0.004325	
R3	20.0433	22.45229	0.004758	
R4	21.52637	23.34758	0.075949	
R5	19.43723	22.45565	0.040287	
R1	18.41338	23.92758	0.006402	
R2	18.34305	22.94143	0.008912	
R3	18.13608	22.43698	0.011331	
R4	19.91082	21.49726	0.06363	

Python parsing script written to perform iterations through results directory. identifying files by established naming conventions. Groups by system and plate, carrying out summary calculations:

1. Average temperature to act as a baseline.
2. Max and Min extremes: acts as identifiers for hotspots and velocity spikes
3. Pressure and Temperature Change calculations: subtracting monitor and outlet conditions from set inlet values.

	A	B	C	D	E	F	G	H
1		Avg Temp	Max Temp	Min Temp	Pressure d	Max velocity		
2	Full system	38.08998	41.62939		19.473742	0.1857376		
3	Plate	30.71276	33.00408	26.93524				
4								



1D Heat Transfer Calculations

DE2 Thermofluids - Part 2

1. Total Heat Load

$$Q = q''' \times L \times W \times H = 103,113 \times 0.42 \times 0.21 \times 0.07 = \mathbf{637\text{ W}}$$

We assume all 637 W transfers through the cold plate.
Ambient convection (~20 W) is negligible and ignored.

2. Flow Regime (3 Diameters)

$$A_c = \frac{\pi D^2}{4}$$

Cross-sectional areas for each diameter

6mm:	2.827×10 ⁻⁵
8mm:	5.027×10 ⁻⁵
10mm:	7.854×10 ⁻⁵

$$V = \frac{\dot{m}}{\rho A_c}$$
$$V = \frac{0.01}{997 \times 5.027 \times 10^{-5}} = 0.200\text{ m/s}$$

Mean Velocity

6mm:	0.355 m/s
8mm:	0.2 m/s
10mm:	0.128 m/s

$$Re = \frac{\rho V D}{\mu}$$
$$Re = \frac{997 \times 0.355 \times 0.006}{0.00089} = 2,384$$

Reynolds Numbers

6mm:	2384 - Bordering Transitional
8mm:	1789 - Laminar
10mm:	1432 - Laminar

3. Nusselt Number + Enhancement Factors

For 6mm, Reynolds number is bordering transitional flow giving us a **Nusselt number of 14.5**. Intermittent turbulent bursts begin to disrupt the laminar boundary layer, significantly enhancing heat transfer beyond the theoretical laminar limit of 4.36.

For both 8mm and 10mm Reynolds numbers remain < 2300, flow regime is laminar. Giving a **Nusselt number of 4.36** for fully developed laminar flow under constant heat flux

$$Nu_{eff} = \varphi \times Nu_{base}$$

$$Nu_{eff} = 2.2 \times 4.36 = 9.59$$
$$Nu_{eff} = 2.2 \times 14.5 = 31.9$$

Nu chosen to be 14.5 due to its transitional state at 6mm diameters.

4. Convection

$$h_{conv} = \frac{Nu_{eff} \times K_{water}}{D} = \frac{9.59 \times 0.6}{0.008} = 719\text{ W/m}^2\text{K}$$

Heat Transfer Coefficients

6mm:	3190
8mm:	719
10mm:	719

$$A_{wetted} = \pi D L_{total}$$

$$A_{wetted} = \pi \times 0.006 \times 0.861 = 0.0162\text{ m}^2$$

Lengths measured point to point, simplified to remove curved circumferences

Total Lengths (estimated over entire cooling plate structure)

6mm:	861.48	6mm:	0.0162m2
8mm:	1554.96	8mm:	0.0391 m2
10mm:	1337.22	10mm:	0.0420 m2

Wetted areas summed = 0.0973 m2

$$h_{convection} = \frac{Nu \times k_{water}}{D} = \frac{9.592 \times 0.6}{0.008} = 719.4$$

$$= \frac{9.592 \times 0.6}{0.010} = 575.5\text{ W/m}^2\text{K}$$

$$= \frac{31.9 \times 0.6}{0.006} = 3190$$

$$R_{conv,6mm} = \frac{1}{h \times A_{wetted}} = \frac{1}{3190 \times 0.0162} = 0.01935\text{ K/W}$$

$$= \frac{1}{719.4 \times 0.0391} = 0.03554\text{ K/W}$$

$$= \frac{1}{575.5 \times 0.0420} = 0.04137\text{ K/W}$$

$$\frac{1}{R_{conv,total}} = \frac{1}{R_{conv,6mm}} + \frac{1}{R_{conv,8mm}} + \frac{1}{R_{conv,10mm}}$$

$$R_{conv,total} = \frac{1}{103.99} = 0.00962\text{ K/W}$$

5. Conduction

Thermal resistance from conduction between bottom area upon cold plate.

$$R_{cond,plate} = \frac{thicknes/2}{K_p \times A_{contact}}$$

$$A_{contact} = L \times W = 0.42 \times 0.21 = 0.0882\text{ m}^2$$

$$R_{cond,plate} = \frac{0.003}{50 \times 0.0882} = 0.00068\text{ K/W}$$

6. Total Thermal Resistances. + Temp. Change

$$R_{total} = R_{conv,total} + R_{cond,plate}$$

$$R_{total} = 0.00962 + 0.00068 = 0.01030\text{ K/W}$$

Difference between middle of cold plate and bottom surface of the battery

$$\Delta T_{plate \rightarrow coolant} = Q \times R_{total} = 637 \times 0.01030 = 6.56\text{ }^{\circ}\text{C}$$

7. Coolent outlet temperature

$$T_{out} = T_{in} + \frac{Q}{\dot{m} c_p} = 20 + \frac{637}{0.01 \times 4180} = 20 + 15.2 = 35.2\text{ }^{\circ}\text{C}$$

$$T_{avg,coolant} = \frac{T_{out} + T_{in}}{2} = \frac{35.2 + 20}{2} = 27.6\text{ }^{\circ}\text{C}$$

8. Battery Surface Temp.

outlet temp.

$$T_{surface} = T_{avg,coolant} + \Delta T_{plate \rightarrow coolant} = 27.6 + 6.56 = 34.16\text{ }^{\circ}\text{C}$$

Temperature differences between middle and bottom of the battery

$$R_{cond,battery} = \frac{H/2}{K_b \times A_{contact}} = \frac{0.035}{30 \times 0.0882} = 0.01323\text{ K/W}$$

Final Temp. Estimation:

42.59°C

Falls within an expected range...reasonable good, will be confirmed and contrastesd agasint cfd results...this is a serious estimation

Pressure Loss Calculations

DE2 Thermofluids - Part 2

Methodology

Due to the branching network topology, mass flow rate, velocity and friction factor vary at every diameter transition and split point along the selected path.

A single-leg hand calculation was attempted but produced $\Delta P = 9.8$ Pa demonstrating the oversimplification is too large to be acceptable. Instead a parameterised python script has been implemented, script uses core pressure loss principles to calculate pressure loss accross a specified path through the system.

[SINGLE LEG MODEL →](#)

[PYTHON PARAMETERISATION](#)

structure of items in section array for path:
`{"d": D3, "L": 0.0312, "split": 1/2, "parallel": False},`

structure of items in junction/bends array for path:

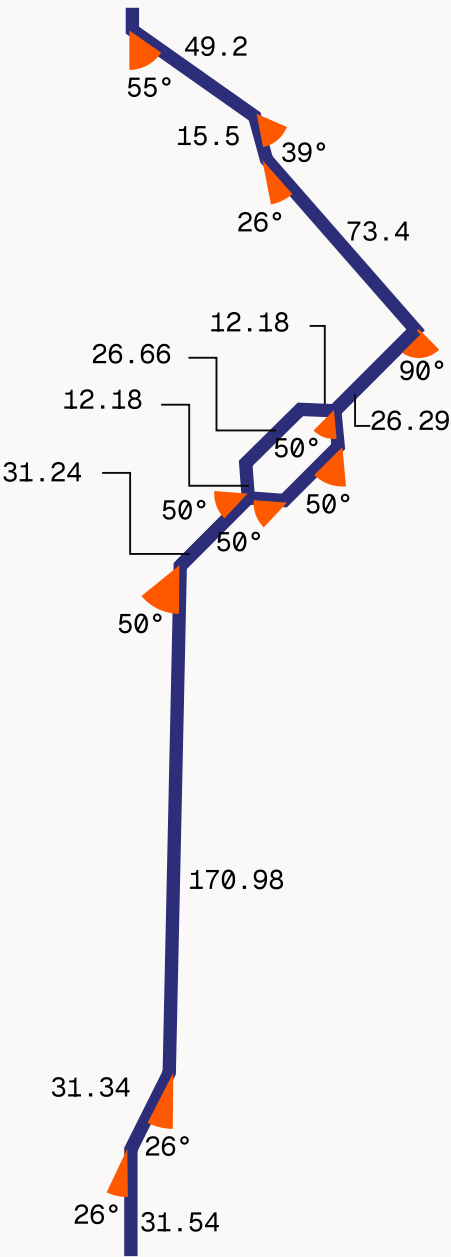
`{"angle": 55, "section": 0},`

Selected path pressure drop:
Delta = 13.98 Pa

[INITIAL HAND CALCULATIONS](#)

$$\Delta P_f = f \times \frac{L}{D} \times \frac{1}{2} \rho V^2 = \frac{32 \mu V L}{D^2}$$
$$\Delta P_f = \frac{32 (0.001) (0.355) (0.0312)}{0.006^2} = 9.845 \text{ Pa}$$

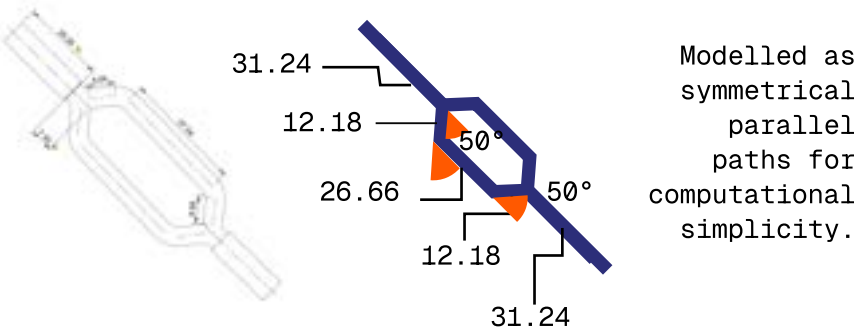
Pressure losses significantly larger then what is realistic or expected... 1D hand calcs struggle to represent complexity



1. Sectional Analysis

Using 1D hand calculations. Setup as 1/10th of whole system inlet flow rate, reflecting central 10 vein structure.

$$\dot{m}_{inlet} = \frac{\dot{m}_{total}}{10} = \frac{0.01}{10} = 0.001 \text{ kg/s at 8mm}$$



equal split between parallel parts: now 0.0005 kg/s

2. Flow Regime (2 Diameters)

$A_c = \frac{\pi D^2}{4}$	Cross-sectional areas for each diameter	6mm: 2.827×10^{-5} 8mm: 5.027×10^{-5}
$V = \frac{\dot{m}}{\rho A_c}$	Flow velocities for each diameter	6mm: 0.01774 m/s 8mm: 0.01996 m/s
$Re = \frac{\rho V D}{\mu}$	Reynolds numbers for each diameter	6mm: 119 8mm: 179
Both Fully Laminar		

3. Friction Factors

$$\text{6mm: } f = \frac{64}{119} = 0.5378 \quad \text{8mm: } f = \frac{64}{179} = 0.3575$$

4. Major Losses

$$\Delta P_f = f \times \frac{L}{D} \times \frac{1}{2} \rho V^2$$
$$\Delta P_f = 0.3575 \times \frac{0.03124}{0.008} \times \frac{1}{2} \times 997 \times 0.01996^2 = 0.2773 \text{ Pa}$$
$$\Delta P_f = 0.5378 \times \frac{0.04056}{0.006} \times \frac{1}{2} \times 997 \times 0.01774^2 = 0.569 \text{ Pa}$$

5. Minor losses

Constriction

$$\Delta P_{contraction} = K_{contraction} = 0.5 \left(1 - \frac{A_2}{A_1} \right)$$
$$K_{contraction} = 0.5 \times \left(1 - \left(\frac{0.006^2}{0.008^2} \right) \right) = 0.2188$$
$$K_{contraction} \times \frac{1}{2} \rho V_6^2 = 0.2188 \times \frac{1}{2} \times 997 \times 0.01774^2 = 0.034 \text{ Pa}$$

50 degree bends

$$\Delta P_{bend} = K_{loss} = \frac{\theta}{90} \times K_{90} = \frac{50}{90} \times 0.493 = 0.2739$$
$$K_{loss} \times \frac{1}{2} \rho V^2 = 0.2739 \times \frac{1}{2} \times 997 \times 0.01774^2 = 0.043 \text{ Pa}$$

x2 for each parallel section

Expansion

$$\Delta P_{expansion} = K_{expansion} = \left(1 - \frac{A_1}{A_2} \right)^2 = (1 - 0.5625)^2 = 0.1914$$
$$K_{expansion} \times \frac{1}{2} \rho V_6^2 = 0.1914 \times \frac{1}{2} \times 997 \times 0.01774^2 = 0.030 \text{ Pa}$$

Splits + Joins

$$\Delta P_{split} = K_{split} \times \frac{1}{2} \rho V_6^2 = 1.0 \times \frac{1}{2} \times 997 \times 0.01774^2 = 0.157 \text{ Pa}$$

(same for joins)

6. Total Losses

$$\Delta P_{inletOutlet} = \Delta P_{contraction} + \Delta P_{expansion} + \Delta P_{split} + \Delta P_{merge}$$
$$= 0.034 + 0.030 + 0.157 + 0.157 = 0.378 \text{ Pa}$$

$$\Delta P_{parallel} = \Delta P_f + 2(\Delta P_{bend}) = 0.569 + 0.086 = 0.655 \text{ Pa per path}$$

equal split and the two paths combine as equal parallel resistors

$$R_A = R_B = \frac{\Delta P_{parallel}}{\dot{m}_{path}} = \frac{0.655}{0.0005} = 1310 \text{ Pa}\cdot\text{s/kg}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_A} + \frac{1}{R_B} = \frac{2}{1310} \implies R_{eq} = 655 \text{ Pa}\cdot\text{s/kg}$$

$$\Delta P_{parallel,eq} = R_{eq} \times \dot{m}_{inlet} = 655 \times 0.001 = 0.655 \text{ Pa}$$

$$\Delta P_{total} = \Delta P_{inletOutlet} + \Delta P_{parallel,eq} = 0.378 + 0.655 = 1.033 \text{ Pa}$$

CFD Results

DE2 Thermofluids - Part 2

Heat Transfer

Maximum, minimum and average temperatures across the entire battery.

T_{min}

29.8°C

T_{avg}

38.08°C

T_{max}

41.629°C

The battery is extremely effective at cooling the battery. The average temperature of 38° is within the 25-40°C range required for optimum battery use.

The temperature range is 10.16°C which is much larger than a battery cooling system should ideally have since it shouldn't be more than 5°

Pressure Drop

The pressure drop was slightly higher

ΔP

19.474 Pa

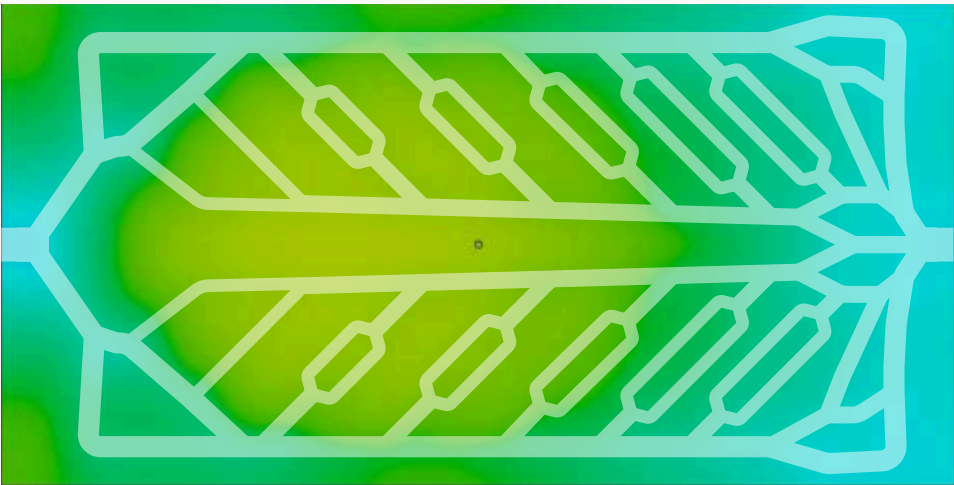
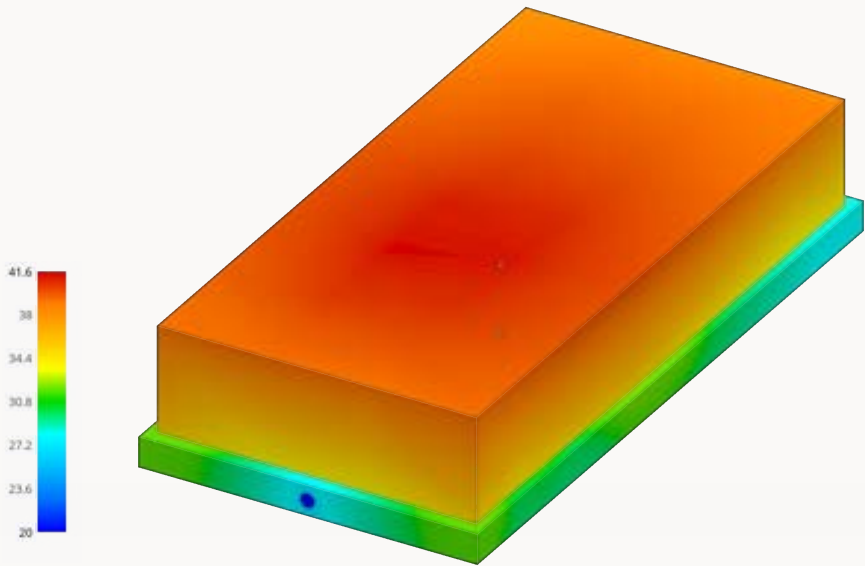
V_{max}

0.1857°C

Heat Transfer

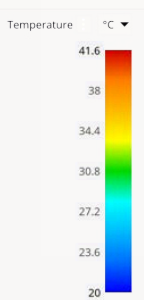
Top of Battery Temperature

Overall, the battery cooling is very even across the entire battery. However, where you would expect the cooling to get worse further down the battery, it improves, with the hotspot shifted towards the inlet instead of the outlet.



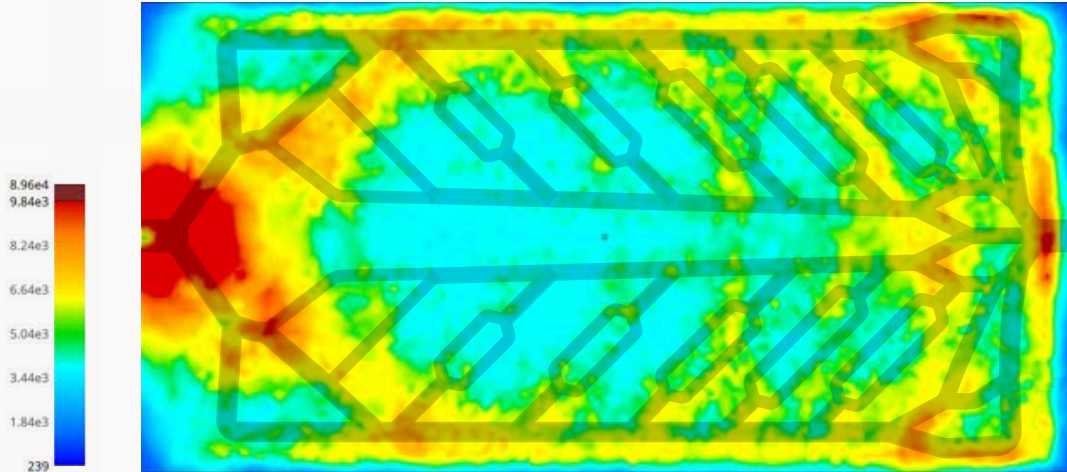
Bottom of Battery Temperature

Viewing the same battery temperatures but from the bottom shows more clearly how the hotter area (where it is more yellow) is further towards the inlet. This suggests that there is not enough water flowing few the top few channels.



Heat Flux Map

Heat flux is the heat flow per unit area in W/m2. Higher values shown in red are where the heat transfer is very strong, cooling more effectively. At the inlet this is clear in the red spot where the cold water flowing in has as high heat flow. The majority of the water flowing around the outside of the pipe is also clear from the higher values on the edges than the middle. The split into three 10mm pipes at the outlet results in a high heat flux showing the need for developing the design with adapted diameters to create a more even flow.



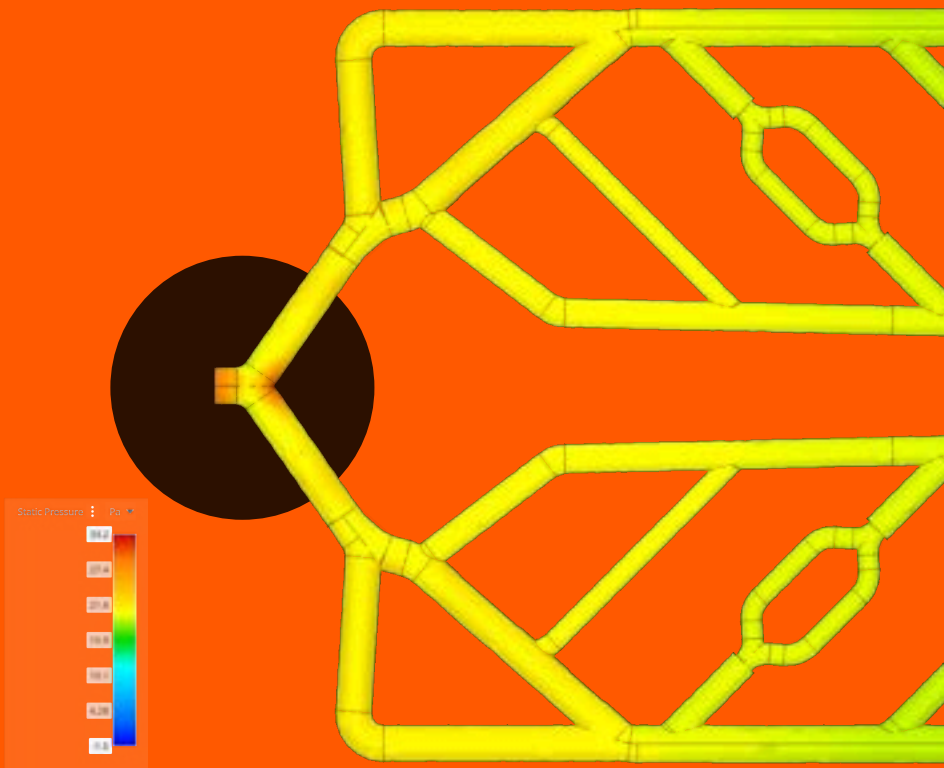
CFD Results

DE2 Thermofluids - Part 2

Macro analysis of the velocity mapping, as a diagnostic view to find key problem areas. By evaluating the global distribution between the initial inlet split through to the final converging junction, we can pinpoint critical pressure loss junctions where geometry hinders performance.

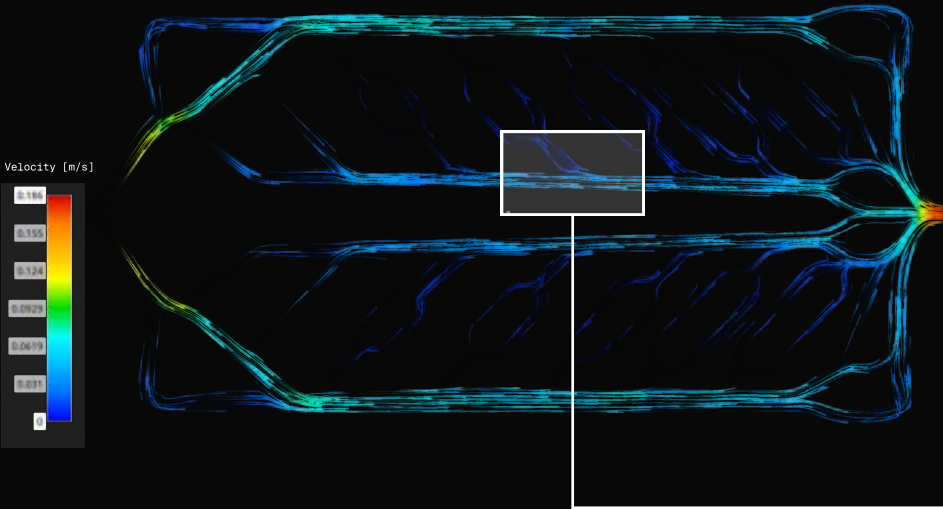
Findings will act as the basis for a micro analysis into internal vein structures and lead us through to determining iterative improvements in battery design.

Static Pressure Plot



Static pressure plot reveals high pressure region at inlet. This is due to the sudden division of flow and sharp geometry. The main concentration of pressure here is the only orange region on the map suggests that it is a significant contributor to the overall system pressure loss. A smoother transition or gradual expansion would reduce the pressure gradient and improve flow distribution.

Velocity Maps

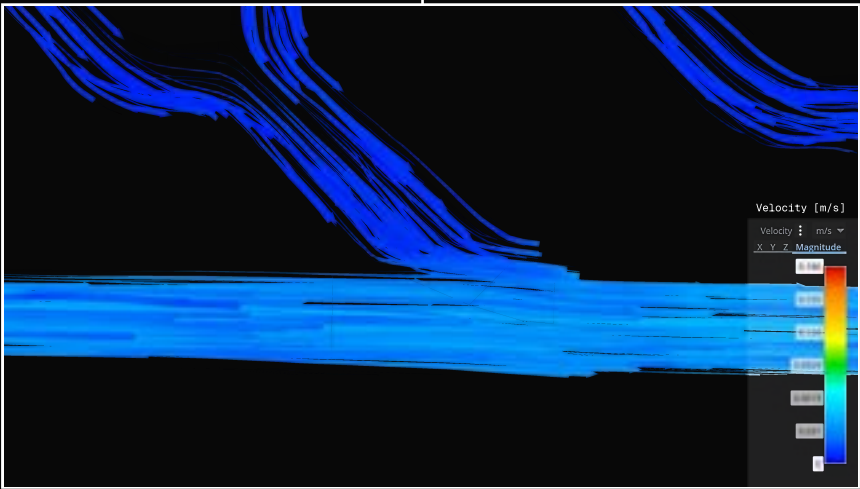


Overall Velocity Map

Wide variety of velocities, higher velocities and more flow lines along the edges of the battery at a faster speed suggests that there is unequal splitting of cooling fluid. Problem especially noticeable in the parallel channels near the inlet suggesting central diameters must be increased so that more fluid moves into the more central area of the pipe. The observations reflect the heat plots - the areas with less heat transfer have less fluid moving through them and at lower speeds.

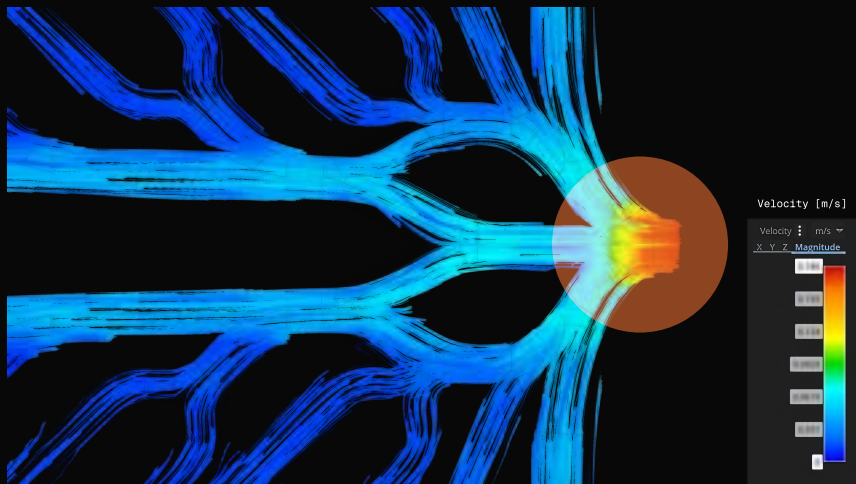
Pipe Split Region

Velocity contours around the bend show flow impinging on the outer wall, creating a high-velocity region and likely flow separation. This indicates that the bend radius is too sharp, preventing the flow from turning smoothly and increasing energy losses. The lack of gradual curvature leads to inefficient flow redirection and higher local pressure drop. Increasing the bend radius would allow smoother acceleration and reduce separation, improving overall performance.



Converging Junction

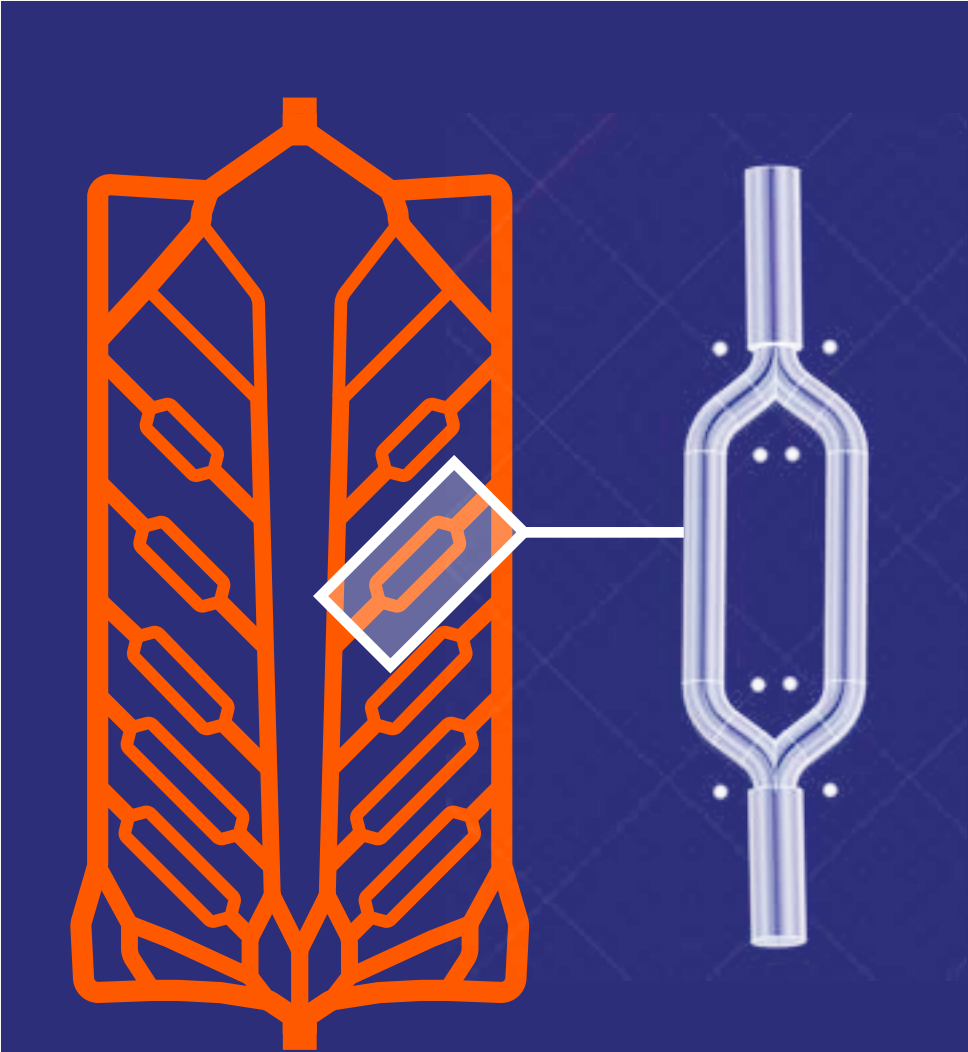
Multiple channels merge into a single region at outlet, creating a strong acceleration and localised high-velocity core. This indicates significant momentum convergence, which leads to elevated pressure losses due to mixing and flow impingement. The abrupt geometry forces the flow to change direction rapidly, causing additional minor losses. A smoother, more gradual manifold design would reduce these effects by allowing a more uniform merging of streams and lowering peak velocities.



INTERNAL VEIN

ZOOM

DE2 Thermofluids - Part 2



We decided to do a more detailed CFD on just one of the internal veins since they repeat ten times throughout the batter plate and therefore their geometry has a significant impact on the overall values.

Using pressure, velocity and temperature plots allowed detailed analysis of every aspect of the geometry to discover areas at which the flow movement was not as expected.

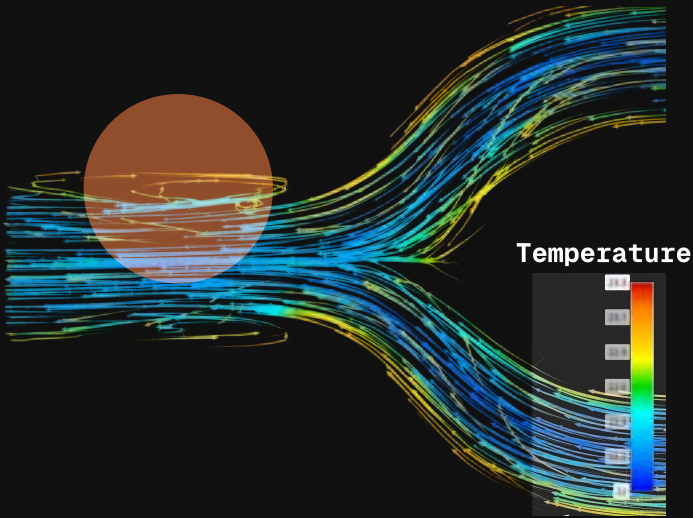
Pressure Drop Comparison

Pressure Distribution and Monitoring

Monitored points show local pressure values that differ slightly from what is perceived from the colour contour alone highlighting importance of extracting quantitative data, as visual gradients can be misleading without precise point measurements.

The colour gradient clearly shows an abrupt pressure drop as the pipe diameter reduces, caused by the corresponding increase in velocity. This behaviour follows continuity and Bernoulli relations:

High-pressure region is observed at the junction where the pipe splits, caused by flow deceleration and redirection. This can be improved by increasing the fillet radius or reducing the turning angle, allowing smoother transition and lowering pressure losses.



Minor Loss - Splitting

Sharp acceleration followed by uneven flow splitting into the two branches at inlet. Abrupt change in direction creates a high-velocity core and regions of recirculation near the inner wall of the bend. This indicates that the geometry forces the flow to turn too quickly, resulting in energy losses due to separation and mixing. Smoother, more gradual transition would help guide the flow and reduce these losses.

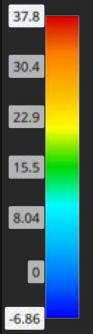
Monitored Points

37.26

20.34

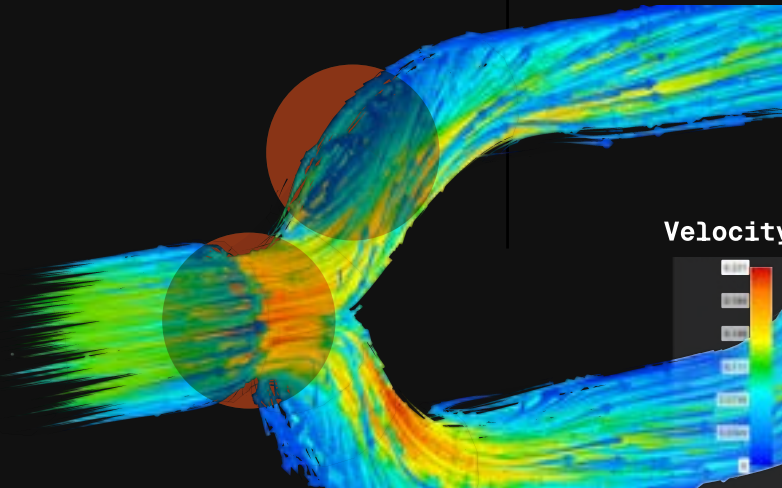
12.97

Pressure Pa



Minor Loss - Expanding Diameter

Clear recirculation zone forming in the expanded section of the channel. Creates a low-velocity region where fluid becomes trapped and circulates rather than progressing downstream. Such recirculation increases losses and reduces effective heat transfer, as slower-moving fluid removes less heat. Issue arises from an overly aggressive expansion angle, which prevents the flow from remaining attached to the wall. A more gradual diffuser angle would reduce separation and improve flow uniformity.



Development

DE2 Thermofluids - Part 2

Based on the observed problems in the CFD, we changed multiple sections, due to the very complex design, geometry changes were not viable in the available time outside of the inlet, we instead focused on changing diameters at key areas to improve the consistency of the cooling.

Maintaining 3 Diameter Structure

Where diameter changes were made within the design, the new values were kept within the existing 3 diameters - 6mm, 8mm and 10mm. This was due to the parameterised system used throughout the process in both the CAD and hand calculation code which allows the values to be easily changed. Due to the structure of the hand calculation python code, the lengths can be changed and set constants for the each diameter used to quickly calculate changes in pressure and heat due to diameter.

The already complex design would be very difficult to manufacture but keeping the number of different pipe diameters to three would keep it from become too expensive while benefitting from potentially better performance.

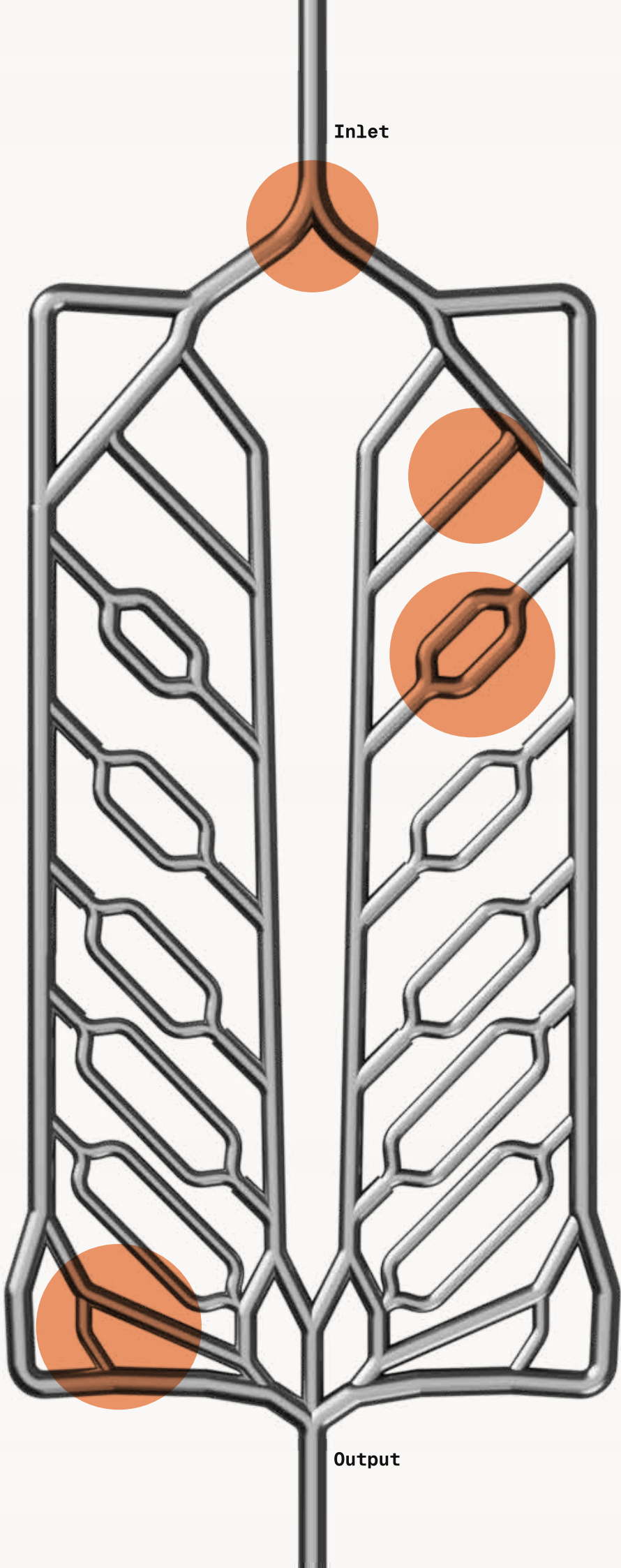
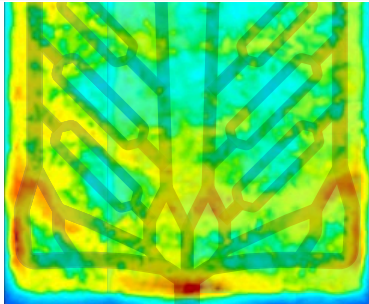
D1=6mm

D2=8mm

D3=10mm

Corners Heat Flux

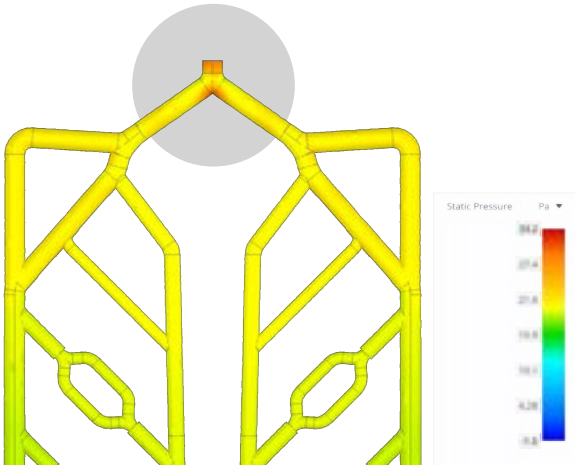
To prevent such a cooling effect on the corners of the battery where it is less useful, pipes in the corners near the outlet had diameters reduced to 6mm instead of 8mm, increasing the resistance of the area so the coolant moves inwards earlier.



Inlet

To reduce the static pressure at the point where the inlet splits into two pipes, the fillet as it passed is increased considerably to a 29mm radius.

The large fillet lets the flow turn gradually so the streamlines stay attached, reducing flow separation. This avoids eddies along the edges as seen in the internal vein zoom where eddies form along the edges of the split. By making the junction less sharp the K value is reduced, reducing the pressure loss across it.

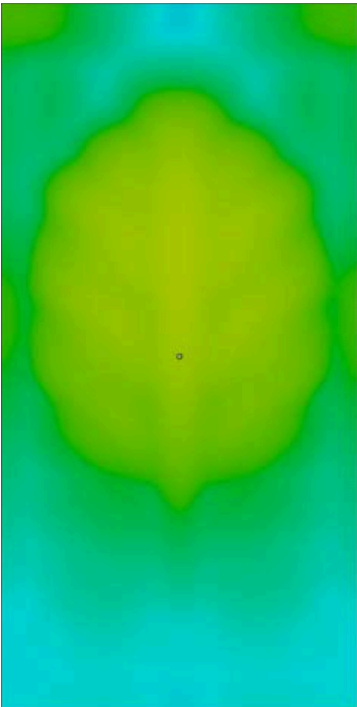


Inner Areas by Inlet

As seen in the heat transfer plots, the battery temperature was hottest slightly towards the inlet instead of the outlet as expected.

This can be amended by finding ways for more heat to be transferred in the central area of the cooling system.

By increasing the diameter of the 6mm top area pipes to 8mm, more water flows into the central region, therefore increasing the speed at which it moves. This in theory increases the heat transfer in the central area since more velocity means more eddies and therefore heat transfer.



Results Discussion

Through studying the monitors and filtering by the largest change it confirms that an overcorrection was made during the second iteration

Overall Results

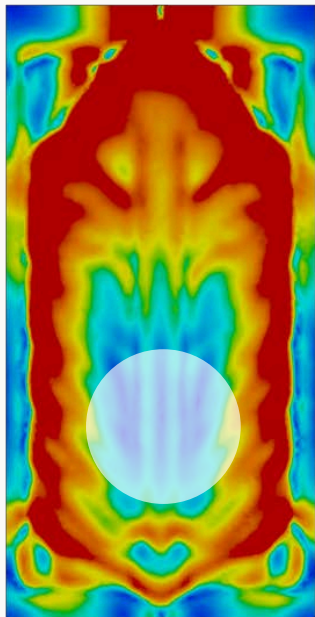
	Max temp	Avg Temp	Min Temp
Old design	42.6	39.2	31.1
New design	50.6	45.9	34.5

	Pressure drop	Max Velocity
Old design	16.6pa	0.185
New design	17.2pa	0.182

Overall the modifications made to our battery cooling geometry were extremely detrimental across all major metrics. After analysis we believe this to be due to an over-correction in dimensions.

The key concern with these results is the increase in the **total temperature difference across the battery of 40%** which is very harmful for battery cells.

The modifications in geometry have also worsened the cooling loop characteristics, increaseing pressure drop while lowering max velocity

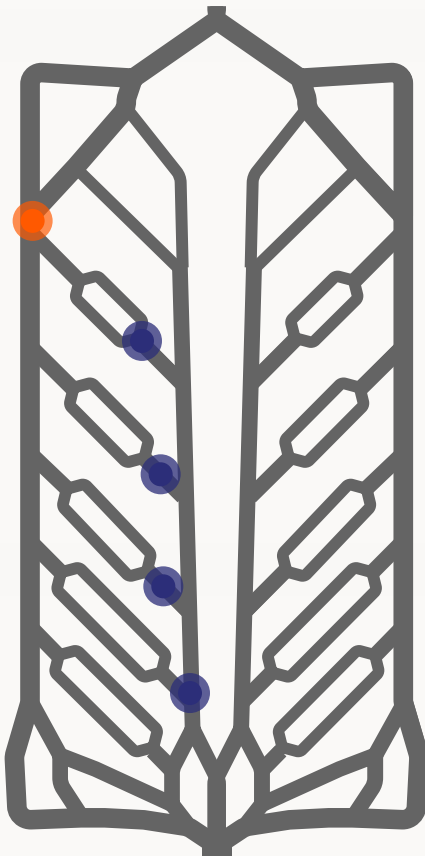


These veins are graphically the largest change between plots, as they become far less effective, with some of the lowest heat flux

The heatflux across the cooling plate appears far higher, this is not due to improved cooling ability across the battery but moreover a visualisation of the increased battery temperature now has a larger gap from

Exploring Monitered Points

Through studying the monitors and filtering by the largest change it confirms that an overcorrection was made during the second iteration



- Here are the top 5 highest increase in temperature monitors. This shows that the change made significantly changes the flow distribution
- Looking at a key point in that is beyond the major modification it's noted that there was a 4.2% decrease in pressure and a 3.5% decrease in velocity. This relatively small change in conjunction with varying branch temperature pressures shows that the deeper cause for a worse performing battery is not merely a bulk flow issue but rather a change in both flow maldistribution and hotter coolant being delivered to downstream points. This attests to the complexity of changes in design with small small geometric changes having a disproportionately negative affect on negative performance

Pressure Loss Calculation Investigation

Investigating the disappointing pressure drop results, the original path pressure loss script was re-run with the updated diameters.

We had constrained to standardised increments of 6mm, 8mm, and 10mm but given the inverse fourth-power relationship between diameter and pressure drop in the laminar regime, these step changes were too coarse.

Investigated results:

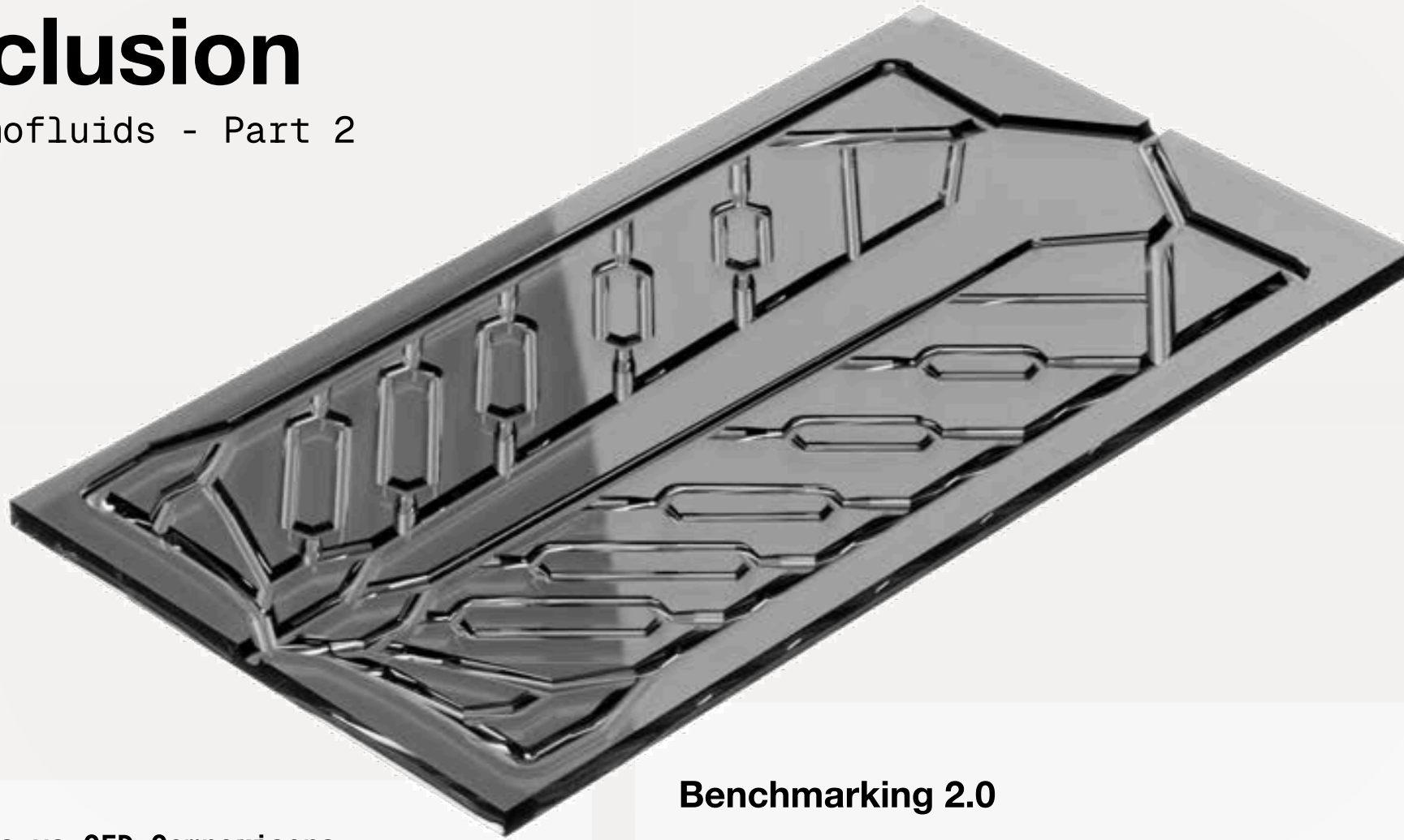
$$\Delta P = \frac{128\mu LQ}{\pi D^4}$$

$\Delta P_{total} = 4.6640 \text{ Pa}$
 $CFD = 19.00 \text{ Pa}$
 $Delta = 14.34 \text{ Pa (75.5\%)}$

Increasing the parallel section diameter draws greater mass flow through the upstream series sections, outweighing the local benefit of the larger diameter as pressure loss of this single path has increase from **13.98 Pa to 14.43 Pa.**

Conclusion

DE2 Thermofluids - Part 2



Hand Calcs vs CFD Comparisons

Temperature Change

42.59°C hand calc result, compared to a CFD Tmax of 41.629°C, this discrepancy of less than 1°C validates the thermal resistance network approach despite its simplifications.

Pressure Drop

A direct comparison between the hand calculation pressure loss of 1.033 Pa over single vein structure and CFD result of 24.29 Pa is not meaningful.

Hand calculation models an isolated, equal-length parallel section using 1/10th of the system inlet flow rate to replicate the vein's embedded position within the wider network. While the CFD simulation on single vein section runs the complete inlet flow.

Benchmarking 2.0

Initial design achieves a pressure drop of 19.5 Pa, sits within the range reported for optimised bio-inspired branching network. Other topology-optimised bio-inspired flow field designs report pressure drops of 6-20 Pa, so our results sit happily at the upper end of this range.

Average temperature falls within the accepted 25-40°C operating window for lithium-ion cells at 38.08 degrees.

Surface temperature range of $\Delta T = 10.16^\circ\text{C}$ fall very short of industry standard of $< 5^\circ\text{C}$
Especially when compared against other bio-inspired network designs which achieve ΔT as low as 3.8°C

Reflection on secondary results

Due to the complexity of the topology and time constraints, a full diameter derivation optimisation was not attempted. Design changes were guided by CFD flow plots and macro heat maps, using visual interpretation of the pressure and velocity distributions to inform where geometric adjustments were needed.

We had constrained to standardised increments of 6mm, 8mm, and 10mm but given the inverse fourth-power relationship between diameter and pressure drop in the laminar regime, these step changes were too coarse.

Redesign Rationale

In hindsight plots should've been used as indicators with numerical data to uphold the foundation of the rationale towards modifications to geometry.

Given more time geometry changes would've been calculated through iterative numerical techniques to find theoretically optimal values, with potentially a small parameter sweep around these values to fine tune.

Appendix - Team Roles

DE2 Thermofluids - Part 2

Team Organisation

The main driver for task delegation was based on personal interest and each member's strengths and excitement toward each stage of the project.

This was then balanced by taking into account time commitments at different stages of both parts of the submission, allowing team members to share the workload while reducing any bottlenecks in the project that could be caused by maldistribution of tasks at any point. We made sure to start early with the CAD and CFD, remembering that the flow of the project involved many aspects which depended on previous steps.

Noah Hett

- Content write-up for Section 1: Car
- Page Plan, layout and formatting for Section 1: Car
- Formatting for Pages 13,14: Hand Calculations

Leila Ali

- CAD development across multiple stages, modelling approaches and iterations to ensure a stable parameterizable model that reliably meshes
- Page formatting and findings write up across all battery pages (excluding hand calculations)

Rhys Wijeratne

- CFD wind tunnel analysis including; method for motion analysis, wind tunnel numerical data handling/plotting
- Vehicle design optimisation content
- CFD for both battery iterations
- Battery CFD results analysis including plots and monitoring results

Eliza-Mae Garvey

- Hand Calculations for both heat transfer and pressure loss (including development of a python script for automation)
- CAD model and CFD model parameterisation and monitoring rationale
- CFD battery analysis

Appendix - References

DE2 Thermofluids - Part 2

REFERENCES - PART 1

REFERENCES - PART 2

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Appendix - Pressure Calc. Python Script

DE2 Thermofluids - Part 2

```
import math

# --- CONSTANTS -----

RHO      = 997          # Density (kg/m³)
MU        = 0.00089     # Dynamic viscosity (Pa·s)
K_FLUID  = 0.6          # Thermal conductivity of water (W/m·K)
PR        = 6.62        # Prandtl number
CP        = 4180        # Specific heat (J/kg·K)
SIGMA     = 5.67051e-8  # Stefan-Boltzmann constant

K_90_SMOOTH = 0.493     # Idelchik Diagram 6-1, smooth 90° bend
CFD_DP      = 19.0      # CFD reference pressure drop (Pa)

# --- PIPE DIAMETERS -----

D1 = 0.010 # m - 10mm
D2 = 0.008 # m - 8mm
D3 = 0.006 # m - 6mm

# --- FLOW PROPERTIES -----

def get_flow_properties(m_dot, d):
    area      = math.pi * (d / 2)**2
    velocity  = m_dot / (RHO * area)
    re        = (RHO * velocity * d) / MU
    return {"velocity": velocity, "re": re, "area_cs": area}

# --- PRESSURE DROP -----

def major_loss(m_dot, d, l):
    """Darcy-Weisbach major loss: dP = 32μVL / D²"""
    props = get_flow_properties(m_dot, d)
    return (32 * MU * props["velocity"] * l) / (d**2)

def minor_loss_bend(m_dot, d, theta_deg, k_90=K_90_SMOOTH):
    """Idelchik Diagram 6-1: K_loss = (theta/90) * K_90, dP = K * 0.5 * rho * V²"""
    props = get_flow_properties(m_dot, d)
    k_loss = (theta_deg / 90.0) * k_90
    dP      = k_loss * 0.5 * RHO * props["velocity"]**2
    return dP, k_loss

# --- PATH ANALYSIS -----
# Each section: diameter (m), length (m), split fraction, parallel flag
# Each bend:    angle (deg), section index it sits at end of

SECTIONS = [
    {"d": D1, "L": 0.0492, "split": 1/2, "parallel": False}, # S1
    {"d": D1, "L": 0.0155, "split": 1/2, "parallel": False}, # S2
    {"d": D1, "L": 0.0734, "split": 1.0, "parallel": False}, # S3
    {"d": D2, "L": 0.02629, "split": 1/2, "parallel": False}, # S4
    {"d": D3, "L": 0.0262, "split": 1/2, "parallel": True }, # S5 – parallel section
    {"d": D2, "L": 0.03124, "split": 1.0, "parallel": False}, # S6
    {"d": D2, "L": 0.171, "split": 1.0, "parallel": False}, # S7
    {"d": D2, "L": 0.03134, "split": 1.0, "parallel": False}, # S8
    {"d": D2, "L": 0.03154, "split": 1.0, "parallel": False}, # S9
]

BENDS = [
    {"angle": 55, "section": 0}, # B1 – end of S1
    {"angle": 39, "section": 1}, # B2 – end of S2
    {"angle": 26, "section": 2}, # B3 – end of S3
    {"angle": 90, "section": 3}, # B4 – end of S4
    {"angle": 50, "section": 4}, # B5 – end of S5 (parallel)
    {"angle": 50, "section": 5}, # B6 – end of S6
    {"angle": 50, "section": 6}, # B7 – end of S7
    {"angle": 50, "section": 7}, # B8 – end of S8
    {"angle": 50, "section": 8}, # B9 – end of S9
    {"angle": 26, "section": 9}, # B10 – end of S10
    {"angle": 26, "section": 9}, # B11 – end of S10
]

def path_pressure_loss(m_total, sections, bends):
    """
    Steps through each section of the selected path.
    Applies correct split mass flow rate at each stage.
    Returns per-section results and totals split by series/parallel.
    """
    m_upstream = m_total
    results     = []

    # Group bends by section index
    bend_map = {}
    for b in bends:
        bend_map.setdefault(b["section"], []).append(b["angle"])
```

```
for i, s in enumerate(sections):
    m_sec = m_upstream * s["split"]
    dP_f  = major_loss(m_sec, s["d"], s["L"])
    dP_c  = sum(minor_loss_bend(m_sec, s["d"], theta)[0]
                for theta in bend_map.get(i, []))

    results.append({
        "d_mm": s["d"] * 1000,
        "L_mm": s["L"] * 1000,
        "m_dot": m_sec,
        "dP_f": dP_f,
        "dP_c": dP_c,
        "parallel": s["parallel"],
    })
    m_upstream = m_sec

dP_f_series = sum(r["dP_f"] for r in results if not r["parallel"])
dP_c_series = sum(r["dP_c"] for r in results if not r["parallel"])
dP_f_parallel = sum(r["dP_f"] for r in results if r["parallel"])
dP_c_parallel = sum(r["dP_c"] for r in results if r["parallel"])
dP_total      = dP_f_series + dP_c_series + dP_f_parallel + dP_c_parallel

return {
    "sections": results,
    "dP_f_series": dP_f_series,
    "dP_c_series": dP_c_series,
    "dP_f_parallel": dP_f_parallel,
    "dP_c_parallel": dP_c_parallel,
    "dP_total": dP_total,
}

# --- HEAT TRANSFER -----

def calculate_h_coeff(re, d):
    nu = 4.36 if re < 2300 else 0.023 * (re**0.8) * (PR**0.4)
    return (nu * K_FLUID) / d

def thermal_analysis(m_dot, d, length, t_surface, t_fluid):
    props = get_flow_properties(m_dot, d)
    h      = calculate_h_coeff(props["re"], d)
    area   = math.pi * d * length
    q      = h * area * (t_surface - t_fluid)
    return {"h": h, "re": props["re"], "heat_removed_w": q}

def radiation_loss(area, epsilon, t_s, t_surr):
    return epsilon * SIGMA * (t_s**4 - t_surr**4) * area

# --- EXECUTION -----

m_total      = 0.01
d_branch     = 0.006

# Heat analysis
thermal = thermal_analysis(m_total / 6, d_branch, 0.25, 318.15, 293.15)
print(f"Heat Analysis (d={d_branch*1000:.0f}mm):")
print(f" Re           : {thermal['re']:.0f}")
print(f" h             : {thermal['h']:.2f} W/m²·K")
print(f" Heat Removed  : {thermal['heat_removed_w']:.2f} W")

# Pressure path analysis
path = path_pressure_loss(m_total, SECTIONS, BENDS)

print(f"\nPressure Loss Path Analysis:")
print(f"  {'Sec':<5} {'D(mm)':>6} {'L(mm)':>8} {'ṁ(kg/s)':>10} "
      f"{'ΔPf(Pa)':>10} {'ΔPc(Pa)':>10} Note")
print(f"  {'-' * 65}")
for i, r in enumerate(path["sections"]):
    flag = "← parallel" if r["parallel"] else ""
    print(f"    S{i+1:<4} {r['d_mm']:>6.0f} {r['L_mm']:>8.2f} {r['m_dot']:>10.6f} "
          f"{r['dP_f']:>10.4f} {r['dP_c']:>10.4f} {flag}")

print(f"\n Series    : ΔP_f = {path['dP_f_series']:.4f} Pa "
      f"ΔP_c = {path['dP_c_series']:.4f} Pa")
print(f" Parallel  : ΔP_f = {path['dP_f_parallel']:.4f} Pa "
      f"ΔP_c = {path['dP_c_parallel']:.4f} Pa")
print(f"\n ΔP_total = {path['dP_total']:.4f} Pa")
print(f" CFD      = {CFD_DP:.2f} Pa")
delta = abs(path["dP_total"] - CFD_DP)
print(f" Delta    = {delta:.2f} Pa  ({delta/CFD_DP*100:.1f}%)")
```